

10

MEASUREMENT AND SPACE

GEOMETRY AND NETWORKS

The word ‘geometry’ comes from the Greek word *geometria*, which means ‘earth measuring’. The principles and ideas of geometry and networks are evident in many aspects of our lives. Geometry can be seen in the design of buildings, bridges, roads and transport networks, such as this footbridge over the Yarra River in central Melbourne.



Chapter outline

	Working mathematically				
10.01 Line and rotational symmetry*	U	F		R	C
10.02 Transformations on the number plane*	U	F		R	C
10.03 Similar figures	U	F		R	C
10.04 Finding sides in similar figures	U	F	PS	R	
10.05 Networks*	U	F		R	C
10.06 Planar graphs and Euler's formula*	U	F		R	C
10.07 Eulerian trails and circuits*	U	F		R	C

*PATH

Wordbank

edge (of a network) A line that connects vertices (points) together

Eulerian trail A walk in a network in which every edge is used only once

planar graph A connected network (graph) where the edges do not cross (overlap)

scale factor The amount by which a shape has been enlarged or reduced, equal to $\frac{\text{image length}}{\text{original length}}$

similar To have the same shape but not necessarily the same size, an enlargement or reduction (symbol: ~)

transformation Translation (slide), reflection (flip) or rotation (turn)

vertex (of a network) A point representing an item that is connected by lines called edges.

Quiz
Wordbank 10

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (4):

10.04 Finding an unknown side in similar figures
10.05 Networks
10.07 Eulerian trails • Euler's formula for planar graphs

Twig videos (3):

10.03 The history of the golden ratio
10.05 The seven bridges of Konigsberg • Networks: Labyrinths and mazes
10.07 The seven bridges of Konigsberg

Quizzes (5):

- Wordbank 10
- SkillCheck 10
- Mental skills 10
- Language of maths 10
- Test yourself 10

Skillsheets (2):

10.01 Line and rotational symmetry
10.04 Finding sides in similar triangles

Worksheets (10):

10.01 Symmetry • Symmetry of flat shapes
10.02 Transformations • 5 mm grid paper
10.03 A page of similar figures • Enlargements and reductions • Enlarging a logo
10.04 Finding sides in similar figures
10.07 Eulerian trails and circuits
Mind map Geometry and networks
Test yourself 5 mm grid paper

Puzzles (4):

SkillCheck Finding angles
10.03 Cartoon enlargement
10.04 Similar triangles
Language of maths Geometry and networks crossword

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

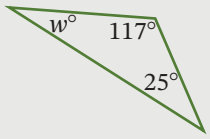
In this chapter you will:

- (PATH) identify line and rotational symmetries in plane shapes
- (PATH) translate, reflect and rotate points and shapes on a number plane using coordinates, where the reflections are across the x- or y- axis and the rotations through multiples of 90° about the origin
- understand the properties of similar figures
- use scale factors to find unknown sides in similar figures
- (PATH) describe networks using appropriate terminology, including vertex, edge, face and degree
- (PATH) describe a planar graph and apply Euler's formula
- (PATH) describe connected graphs, walks, trails, circuits, paths, cycles
- (PATH) identify and investigate Eulerian trails and circuits

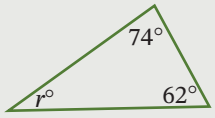
SkillCheck ANSWERS ON P. 623

1 Find the value of each variable.

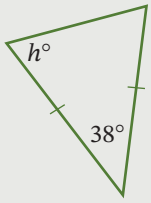
a



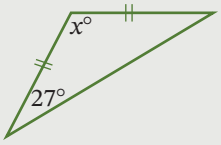
b



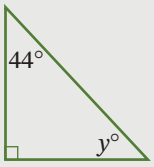
c



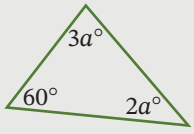
d



e



f



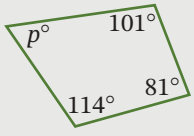
2 Solve each equation.

a $\frac{x}{24} = \frac{5}{8}$ **b** $\frac{m}{7} = \frac{10}{14}$ **c** $\frac{a}{2} = \frac{4}{8}$ **d** $\frac{k}{15} = \frac{2}{3}$

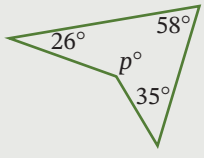
e $\frac{7}{12} = \frac{x}{3}$ **f** $\frac{5}{6} = \frac{n}{4}$ **g** $\frac{t}{2} = \frac{7}{3}$ **h** $\frac{d}{5} = \frac{17}{10}$

3 Find the value of p in each diagram.

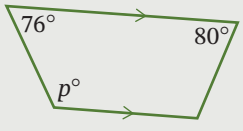
a



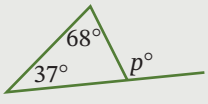
b



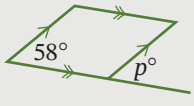
c



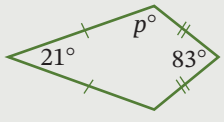
d



e



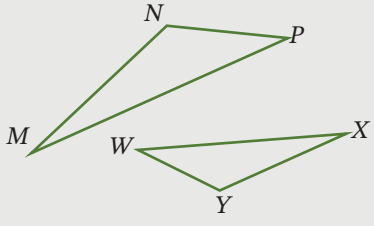
f



4 Triangles MNP and WXY are similar.

a List all pairs of matching angles.

b List all pairs of matching sides.



Quiz
SkillCheck 10

Puzzle
Finding angles

10

10.01 Line and rotational symmetry

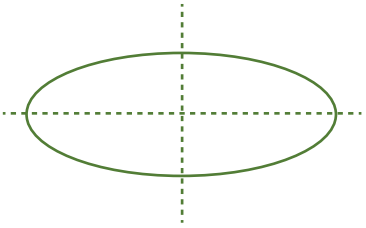
PATH

Skillsheet
Line and rotational symmetry

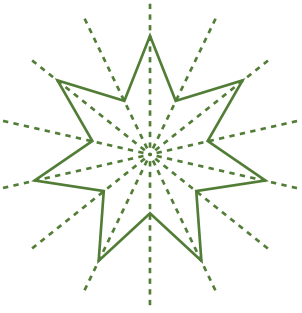
Worksheets
Symmetry

Symmetry of flat shapes

When a shape with **line symmetry** is folded along a line, one half of the shape will fit exactly on top of the other half. The fold line is called the **axis of symmetry**.

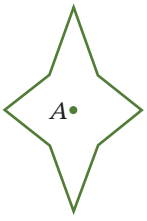


This ellipse has 2 axes of symmetry.

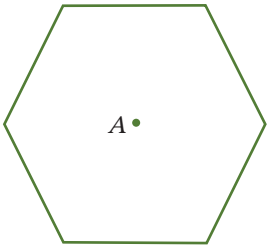


This star shape has 7 axes of symmetry.

When a shape with **rotational symmetry** is rotated about the point, called the **centre of symmetry**, it fits on itself at least once before one complete revolution.



This star shape fits on itself twice in one revolution, so it has rotational symmetry of order 2.



The regular hexagon fits on itself 6 times in one revolution, so it has rotational symmetry of order 6.

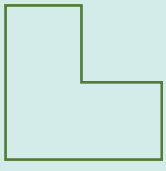
Example 1

For each shape:

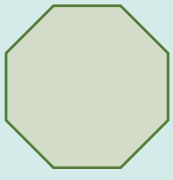
i count how many axes of symmetry it has

ii state whether it has rotational symmetry and the order if it does.

a



b

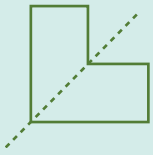


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Solution

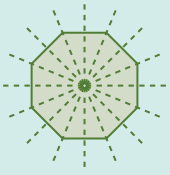
a **i** 1 axis of symmetry

ii no rotational symmetry



b **i** 8 axes of symmetry

ii rotational symmetry of order 8



EXERCISE 10.01 ANSWERS ON P. 623

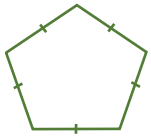
Line and rotational symmetry U F R C

1 Copy each shape and mark its axes of symmetry. R C

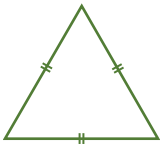
a



b



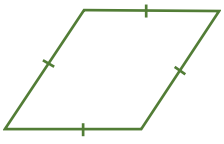
c



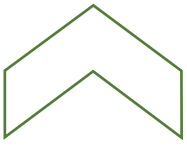
d



e



f



2 State whether each shape has rotational symmetry and the order if it does. R C

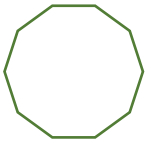
a



b



c



3 Which of these capital letters have: R C

B C H M O P S W X Y Z

a line symmetry? **b** rotational symmetry?

Foundation Standard Complex

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PATH

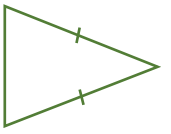
EXAMPLE 1

4 For each shape: R C

i find how many axes of symmetry it has

ii state whether it has rotational symmetry and the order if it does.

a



b



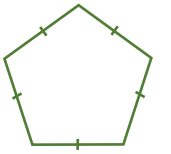
c



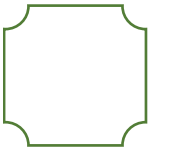
d



e



f



5 For each shape, find: R C

i the number of axes of symmetry

ii the order of rotational symmetry.

a square **b** rhombus **c** regular decagon (10 sides)

d circle **e** kite

6 **a** Find the number of axes of symmetry and the order of rotational symmetry for:

i a scalene triangle **ii** an isosceles triangle **iii** an equilateral triangle.

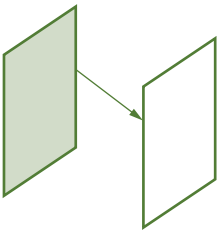
b What do you notice about your answers to part **a**? R C

10.02 Transformations on the number plane

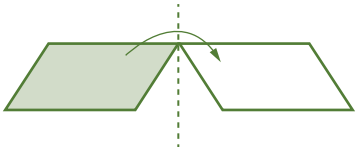
PATH

Worksheet
Transformations

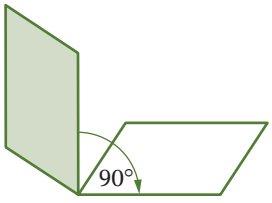
There are 3 types of **transformations**.



translation ('slide')



reflection ('flip')



rotation ('spin')

The shapes have changed position (transformed) but still have the same shape and size.

The original shape is called the **original**.

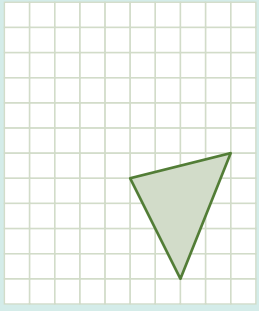
The transformed shape is called the **image**.

Foundation Standard Complex

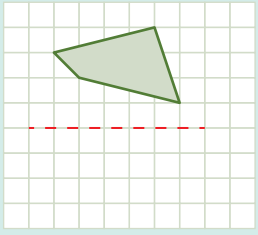
9780170479622

Example 2

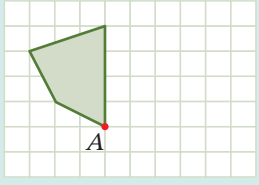
a Translate the triangle 4 units left and 5 units up.



b Reflect this quadrilateral across the dotted line, the **line of reflection**.

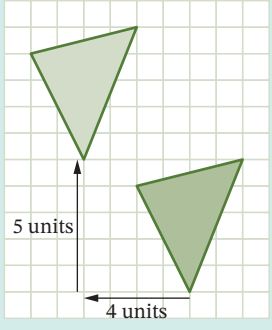


c Rotate the shape 90° in a clockwise direction about the point A, the centre of rotation.



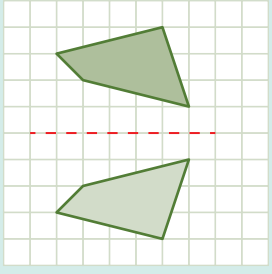
Solution

a



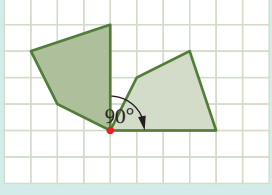
Every point on the original is shifted 4 units left and 5 units up.

b



The original and its mirror image are the same distance from the line of reflection.

c



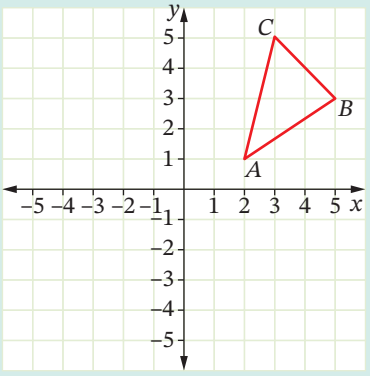
Every point on the original and image are the same distance from the centre of rotation.

PATH

When a point or vertex P of an original shape is transformed, the corresponding point or vertex on the image is labelled P', pronounced 'P-dash' or 'P-prime'.

Example 3

a Reflect $\triangle ABC$ across the y-axis to obtain the image $\triangle A'B'C'$.

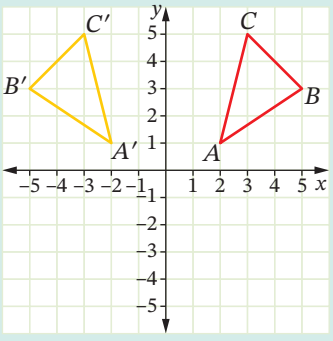


b Compare the coordinates of each vertex of $\triangle ABC$ to its matching vertex in $\triangle A'B'C'$.

c Now reflect $\triangle ABC$ across the x-axis to obtain the image $\triangle A'B'C'$. Compare the coordinates of each vertex of the original to its matching vertex in its image.

Solution

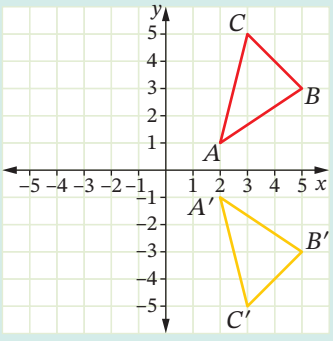
a



b $A(2, 1) \rightarrow A'(-2, 1)$, $B(5, 3) \rightarrow B'(-5, 3)$, $C(3, 5) \rightarrow C'(-3, 5)$

When reflected in the y-axis, the x-coordinate of each vertex changes sign while the y-coordinate stays the same.

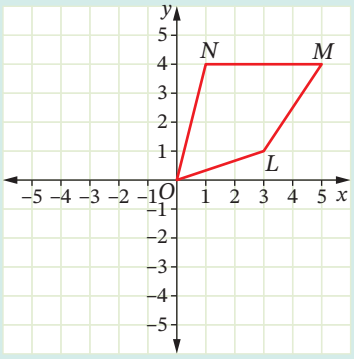
c



$A(2, 1) \rightarrow A'(2, -1)$, $B(5, 3) \rightarrow B'(5, -3)$, $C(3, 5) \rightarrow C'(3, -5)$

When reflected in the x-axis, the y-coordinate of each vertex changes sign while the x-coordinate stays the same.

Example 4



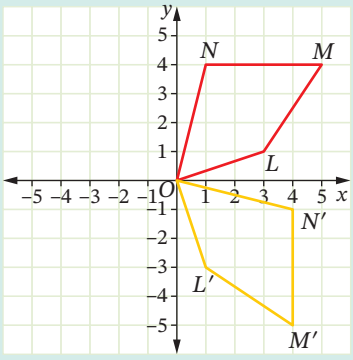
a Rotate $OLMN$ clockwise about O through each angle to obtain $OL'M'N'$:

i 90° **ii** 180° **iii** 270°

b Compare the coordinates of each vertex of $OLMN$ to each vertex of the image $OL'M'N'$ for each of the rotations in part **a**.

Solution

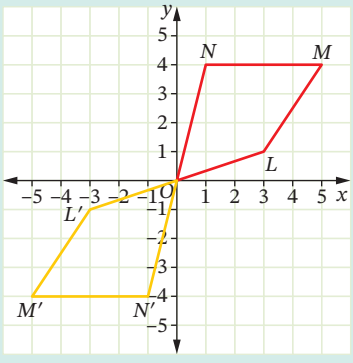
a i



b i $O(0, 0) \rightarrow O(0, 0)$. O is the centre of rotation.

$L(3, 1) \rightarrow L'(1, -3)$,
$M(5, 4) \rightarrow M'(4, -5)$,
$N(1, 4) \rightarrow N'(4, -1)$.

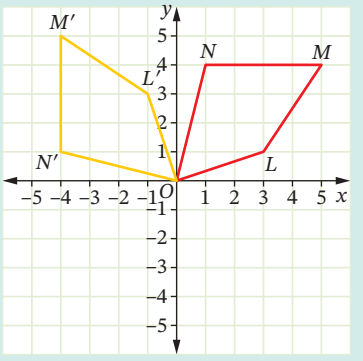
a ii



b ii $O(0, 0) \rightarrow O(0, 0)$,
$L(3, 1) \rightarrow L'(-3, -1)$,
$M(5, 4) \rightarrow M'(-5, -4)$,
$N(1, 4) \rightarrow N'(-1, -4)$.

PATH

a iii



b iii $O(0, 0) \rightarrow O(0, 0)$
$L(3, 1) \rightarrow L'(-1, 3)$,
$M(5, 4) \rightarrow M'(-4, 5)$,
$N(1, 4) \rightarrow N'(-4, 1)$.

Example 5

Find the new position of the point $K(-2, 5)$ if it is:

a rotated about the origin clockwise through 180°

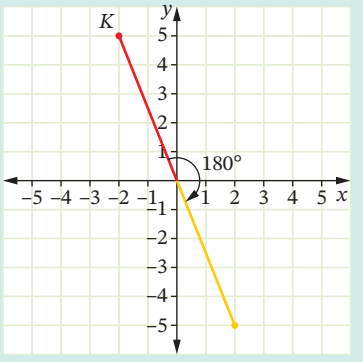
b translated 4 units to the right and 7 units down

c reflected in the y-axis.

Solution

a K is rotated anticlockwise about the origin.

$\therefore$ the new position of K is $(2, -5)$



b The new position of K is $(-2 + 4, 5 - 7) = (2, -2)$.

c K is reflected in the y-axis so the y-coordinate stays the same.

$\therefore$ the new position of K is $(2, 5)$

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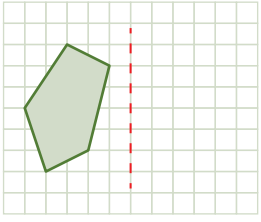
EXERCISE 10.02 ANSWERS ON P. 623

Transformations on the number plane UFRC

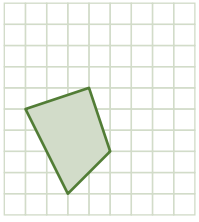
EXAMPLE 2

1 Copy each shape onto grid paper and draw the image after performing the stated transformation.

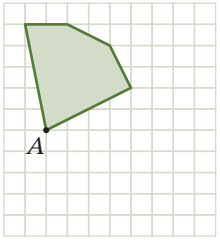
a Reflect across the dotted line



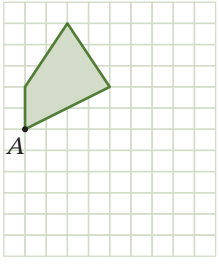
b Translate 3 units right and 3 units up



c Rotate 90° clockwise about A



2 Copy this shape onto grid paper and draw the image of the shape after it has been translated 6 units to the right, 5 units down, followed by an anticlockwise rotation of 90°.

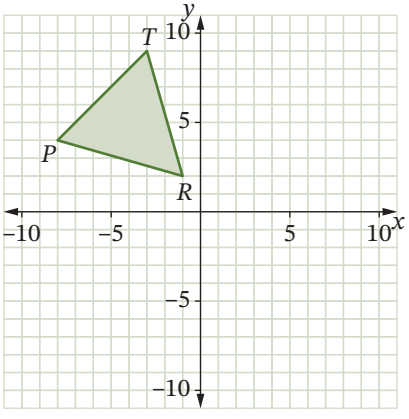


EXAMPLE 3

3 **a** Copy and reflect the shape $\triangle PRT$ across the y-axis to obtain the image $\triangle P'R'T'$.

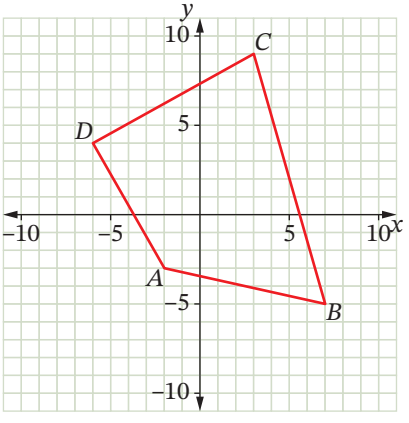
b Compare the coordinates of each vertex of $\triangle PRT$ to its matching vertex in $\triangle P'R'T'$.

c Now copy and reflect $\triangle PRT$ across the x-axis to obtain the image $\triangle P'R'T'$. Compare the coordinates of each vertex of $\triangle P'R'T'$ to its matching vertex in $\triangle PRT$. R C



4 **a** Copy and reflect the quadrilateral $ABCD$ across the x-axis to obtain the image $A'B'C'D'$.

b Write the coordinates of each vertex of $A'B'C'D'$ and compare them to the coordinates of the vertices of $ABCD$. R C



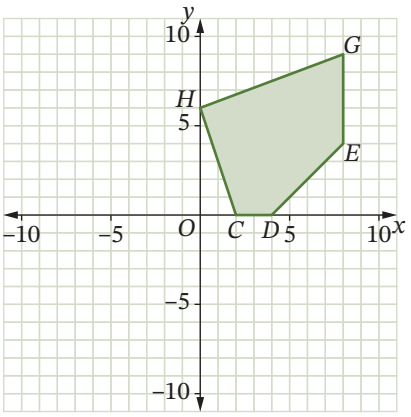
Foundation Standard Complex

Worksheet
5 mm grid paper

PATH

EXAMPLE 4

5 **a** Copy and rotate the shape $CDEGH$ clockwise about O through 90° to create the image $C'D'E'G'H'$.



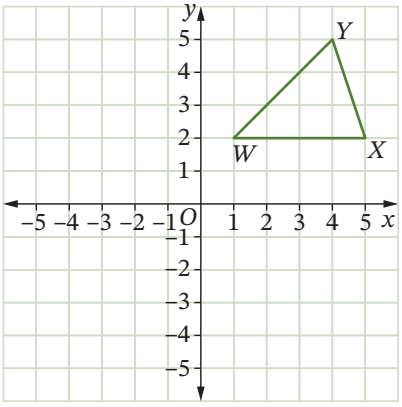
b Write down the coordinates of D' and E' and compare them to the coordinates of D and E.

c Write down the new positions of C, G and H if $CDEGH$ is rotated about O through: R C

i 180° **ii** 270°

6 **a** Copy and rotate $\triangle WXY$ clockwise about O through the following angles to obtain $\triangle W'X'Y'$:

i 90° **ii** 180° **iii** 270°



b Compare the coordinates of the vertices of $\triangle WXY$ to those of $\triangle W'X'Y'$ for each of the rotations. R C

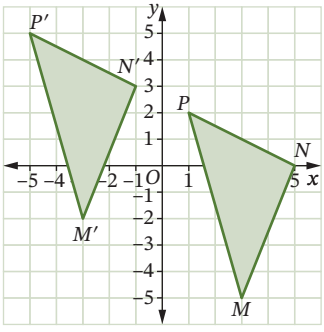
c Copy and rotate $\triangle WXY$ anticlockwise about O through the following angles

i 90° **ii** 180° **iii** 270°

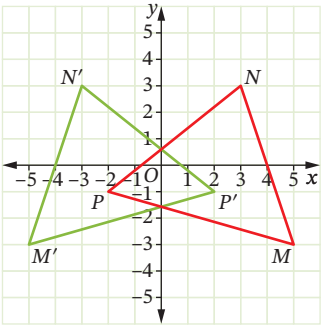
d Which of the images from part **c** are the same as the images from part **a**?

7 Describe the transformations that has been performed on $\triangle MNP$ in each diagram. R C

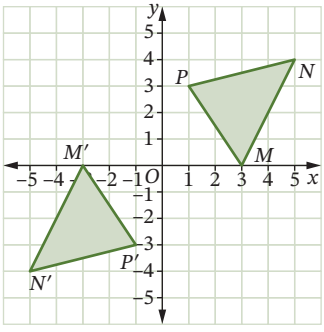
a



b



c

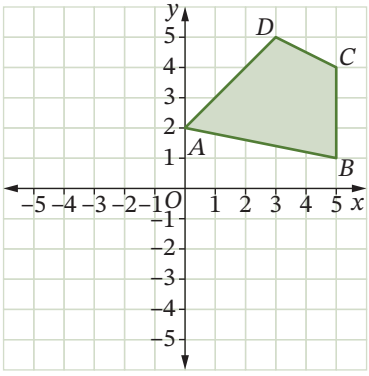


Foundation Standard Complex

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8 **a** Rotate the quadrilateral $ABCD$ about O through 270° and then reflect the image in the x-axis to obtain $A'B'C'D'$.

b Compare the coordinates of the vertices of $ABCD$ to those of $A'B'C'D'$. R C



9 The point $T(4, -3)$ is reflected in the x-axis. What are the new coordinates of T? Select the correct answer **A**, **B**, **C** or **D**.

A $(-4, -3)$ **B** $(-4, 3)$ **C** $(4, 3)$ **D** $(4, -3)$

10 The point $W(-2, -5)$ is rotated through 180° about the origin to obtain the point W'. What are the coordinates of W'? Select **A**, **B**, **C** or **D**.

A $(5, 2)$ **B** $(2, 5)$ **C** $(-5, 2)$ **D** $(-2, 5)$

11 Find the coordinates of the image point A' if $A(-4, -6)$ is:

a reflected in the x-axis

b reflected in the y-axis

c translated 4 units right and 7 units up

d rotated clockwise about the origin through

i 90° **ii** 180° **iii** 270°

12 What is the new position of $D(3, -8)$ if it is reflected in the x-axis and then rotated clockwise about the origin through 90°? Select **A**, **B**, **C** or **D**.

A $(8, -3)$ **B** $(-3, 8)$ **C** $(-8, 3)$ **D** $(-3, -8)$

13 The point $M(5, 2)$ is rotated anticlockwise about the origin through 90° and then reflected in the y-axis. What are the new coordinates of M? Select **A**, **B**, **C** or **D**.

A $(-2, -5)$ **B** $(5, -2)$ **C** $(-5, -2)$ **D** $(2, 5)$

14 Find the new position of $X(-3, -7)$ if it is: R C

a rotated through 180° about the origin

b translated 5 units to the right and 9 units up

c reflected in the x-axis

d rotated clockwise through 90° and then reflected in the y-axis

e translated 6 units to the right, 4 units up and then rotated anticlockwise about the origin through 90°.

15 Find the coordinates of the image point X' if $X(-4, -8)$ has been rotated about the origin through 180° and then reflected in the x-axis. Select **A**, **B**, **C** or **D**.

A $(-4, 8)$ **B** $(4, 8)$ **C** $(8, -4)$ **D** $(4, -8)$

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PATH

16 The point $T(3, -4)$ is the image of a point after it has rotated through 180° and then has been translated 3 units down and 5 units to the right. Find the original point. R C

17 Find the original point if (1, 5) is the image of a point that has been rotated 90° clockwise about the origin and then reflected in the x-axis.

Technology

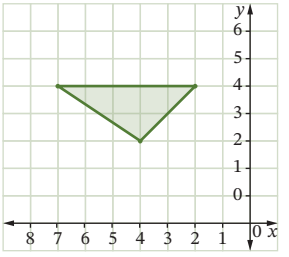
Transformations on the number plane

In this activity you will create reflections, rotations and translations of various shapes using **dynamic geometry software.** Create a number plane that shows all 4 quadrants and values of x and y from –10 to 10.

Activity 1: Reflection

1 Draw a **triangle** with vertices at (–7, 5), (–2, 5) and (–4, 3).

2 Draw a **reflection line on the *y*-axis,** through the points (0, 6) and (0, 0).



3 Reflect the triangle across the line.

4 Click a vertex and drag it. What happens?

5 Draw a new **reflection line** through the points (2, 7) and (–5, –2).

6 Reflect the triangle across this line.

Activity 2: Rotation

1 Draw a **polygon** with vertices at (0, 3), (–2, 3), (–3, 2), (–2, 0) and (0, 3).

2 Select the polygon and rotate it 45° clockwise about the point (0, 3).

3 Repeat step **2**, then rotate the polygon 180° clockwise about (0, 3). What happens?

4 Try other points as the centre point and enter different angles. What happens?

Activity 3: Translation

1 Draw a **polygon** with vertices at (–3, 4), (–3, –1), (1, –1), (1, 1), (–2, 1), (–2, 4) and (–3, 4).

2 Translate the polygon so that the image of (1, 1) is (4, 3).

3 Now translate the shape in 2 other ways.

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Activity 4

1 Construct a **Polygon**.

2 Reflect the polygon about a point.

3 Reflect the polygon about a line.

4 Rotate the polygon by an angle.

5 Translate the polygon by a vector.



Quiz
Mental skills 10

Mental skills 10: Maths without calculators ANSWERS ON P. 625

Dividing a quantity in a given ratio

1 Study this example.

Divide \$5600 between Alice and Peter in the ratio 5 : 3.

Total number of parts = 5 + 3 = 8.

1 part = \$5600 ÷ 8 = \$700

Alice's share = 5 × \$700 = \$3500

Peter's share = 3 × \$700 = \$2100

Check: \$3500 + \$2100 = \$5600 (original amount)

2 Now divide each of these quantities in the given ratio.

a Divide \$150 between Mark and Jenni in the ratio 2 : 1.

b Divide \$2100 between Simon and Sunil in the ratio 4 : 3.

c Divide \$720 between Lisa and Bree in the ratio 2 : 7.

d Divide \$2000 between William and Adriana in the ratio 1 : 3.

e Divide \$4500 between Anne and Pete in the ratio 3 : 2.

f Divide \$3000 between Sharanya and Asam in the ratio 3 : 7.

g Divide \$3600 between Cindy and Carmen in the ratio 5 : 1.

h Divide \$1600 between Nancy and John in the ratio 3 : 5.

i Divide \$990 between Carol and Louis in the ratio 5 : 4.

j Divide \$4000 between Yvette and Andre in the ratio 1 : 4.

k Divide \$4900 between Arden and Ivan in the ratio 3 : 4.

l Divide \$3200 between Tan and Mai in the ratio 5 : 3.

10.03 Similar figures

Video
The history of the golden ratio

Worksheets
A page of similar figures

Enlargements and reductions

Enlarging a logo

Puzzle
Cartoon enlargement

Similar figures have the same shape, but are not necessarily the same size.

When a figure is enlarged or reduced, a **similar figure** is created. The original figure is called the **original**, while the enlarged or reduced figure is called the **image**.



Original Image

The **scale factor** describes by how much a figure has been enlarged or reduced.

Scale factor

$$\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$$

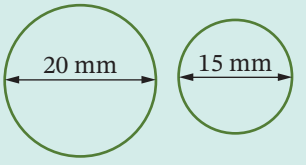
- If the scale factor is greater than 1, then the image is an enlargement.
- If the scale factor is between 0 and 1, then the image is a reduction.

Example 6

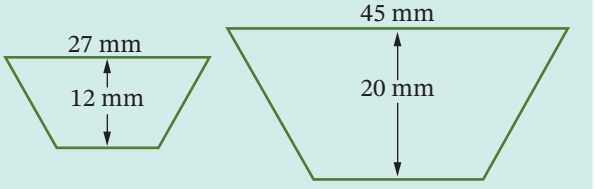
Find the scale factor for each pair of similar figures.

In all questions, assume the left figure is the original and the right figure is the image.

a



b



Solution

a Scale factor $= \frac{15}{20}$ $\qquad \frac{\text{image length}}{\text{original length}}$

$= \frac{3}{4}$

b Scale factor $= \frac{45}{27}$ $\left(\text{or } \frac{20}{12}\right)$ $\qquad \frac{\text{image length}}{\text{original length}}$

$= \frac{5}{3}$

The symbol for 'is similar to' is '~'. When using this notation, we must make sure that the vertices (angles) of similar figures are written in matching order.

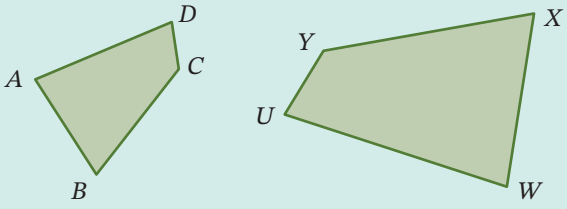
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Properties of similar figures

- Matching angles are equal
- Matching sides are in the same ratio

Example 7

These 2 quadrilaterals $ABCD$ and $UWXY$ are similar.



a List all pairs of matching sides and matching angles.

b Use the correct notation to write a similarity statement relating these 2 quadrilaterals.

Solution

a By rotating the figure $ABCD$, its shape can be matched with $UWXY$.

The pairs of matching sides are:	The pairs of matching angles are:
AB and WX	$\angle A$ and $\angle W$
BC and XY	$\angle B$ and $\angle X$
CD and YU	$\angle C$ and $\angle Y$
AD and WU.	$\angle D$ and $\angle U$.

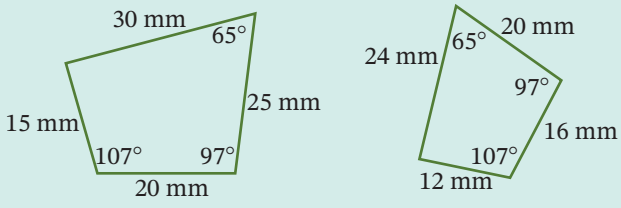
b $\angle A$ matches with $\angle W$, $\angle B$ matches with $\angle X$, $\angle C$ matches with $\angle Y$, $\angle D$ matches with $\angle U$.

$\therefore ABCD \sim WXYU$ Matching order of vertices

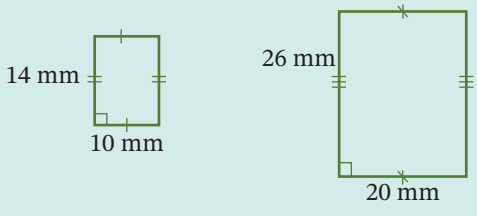
Example 8

Test whether each pair of figures are similar.

a



b



Solution

a For the 2 quadrilaterals, matching angles are equal and the ratios of matching sides are equal.

$\frac{20}{16}=\frac{5}{4}, \frac{25}{20}=\frac{5}{4}, \frac{30}{24}=\frac{5}{4}, \frac{15}{12}=\frac{5}{4}$

$\therefore$ The quadrilaterals are similar.

b For the 2 rectangles, matching angles are equal (90°), but the ratios of matching sides are not equal.

$\frac{10}{20}=\frac{1}{2}$ but $\frac{14}{26}=\frac{7}{13}$

$\therefore$ The rectangles are not similar.

EXERCISE 10.03 ANSWERS ON P. 625

Similar figures U F R C

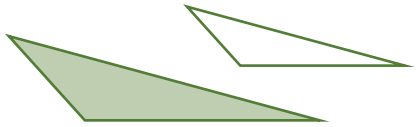
EXAMPLE 6

1 By measurement, find the scale factor for each pair of similar figures.

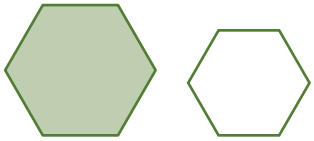
a



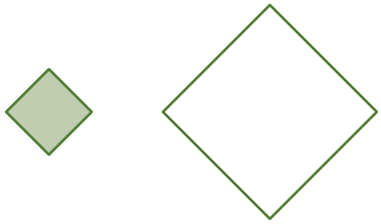
b



c

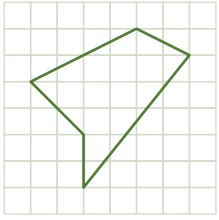


d

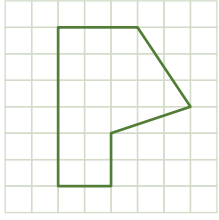


2 Copy each figure onto graph paper and draw its image using the given scale factor.

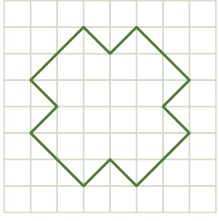
a Scale factor $= \frac{1}{2}$



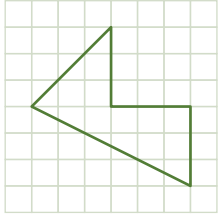
b Scale factor $= 3$



c Scale factor $= 2.5$



d Scale factor $= \frac{2}{3}$



Foundation Standard Complex

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EXAMPLE 7

10.03

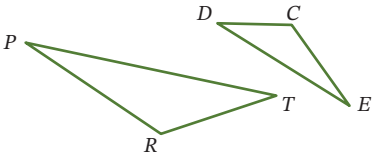
3 For each pair of similar figures: R C

i list all pairs of matching angles

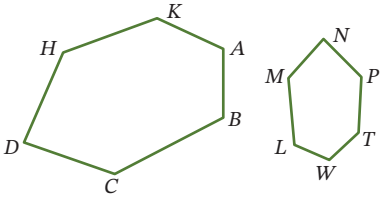
ii list all pairs of matching sides

iii use the correct notation ('~') to write a similarity statement relating them.

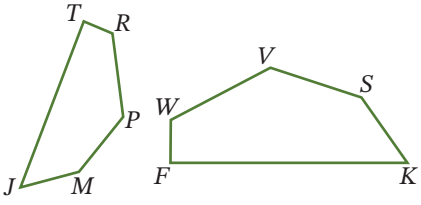
a



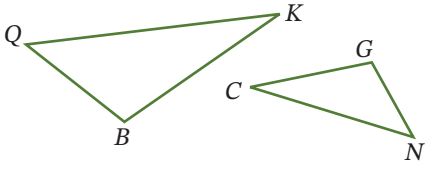
b



c



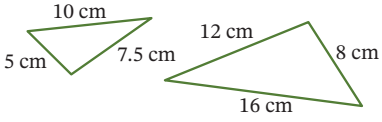
d



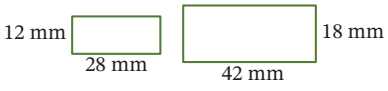
EXAMPLE 8

4 Test whether each pair of figures are similar. R C

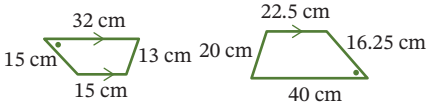
a



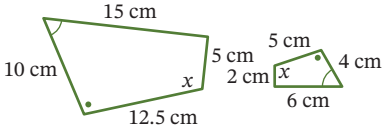
b



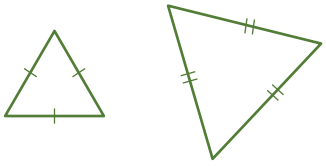
c



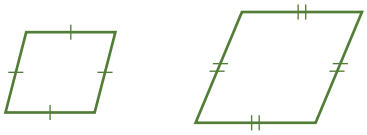
d



e



f



Foundation Standard Complex

10.04 Finding sides in similar figures

Video
Finding an unknown side in similar figures

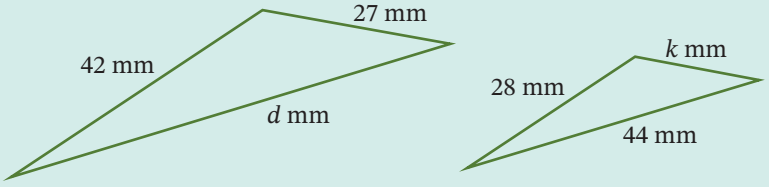
Skillsheet
Finding sides in similar triangles

Worksheet
Finding sides in similar figures

Puzzle
Similar triangles

Example 9

The 2 triangles are similar. Find the values of d and k.



Solution

Since the triangles are similar, the ratios of matching sides are equal.

$\frac{d}{44} = \frac{42}{28}$

$d = \frac{42}{28} \times 44$

$= 66$

$\frac{k}{27} = \frac{28}{42}$

$k = \frac{28}{42} \times 27$

$= 18$

Alternative method:

Scale factor $= \frac{28}{42} = \frac{2}{3}$

$d = 44 \div \frac{2}{3}$

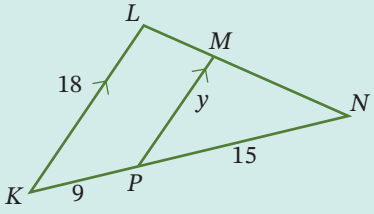
$= 66$

$k = 27 \times \frac{2}{3}$

$= 18$

Example 10

$\triangle KLN \sim \triangle PMN$. Find the value of y.



Solution

$\frac{MP}{LK} = \frac{PN}{KN}$ Ratios of matching sides are equal.

$\frac{y}{18} = \frac{15}{24}$ $KN = 9 + 15 = 24$

$y = \frac{15}{24} \times 18$

$= 11\frac{1}{4}$

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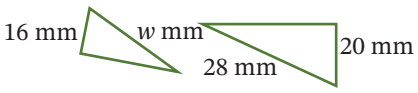
EXERCISE 10.04 ANSWERS ON P. 626

Finding sides in similar figures U F PS R

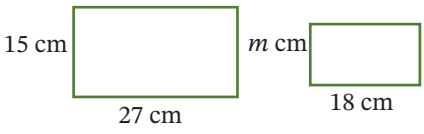
EXAMPLE 9

1 Find the value of each variable in each pair of similar figures. R

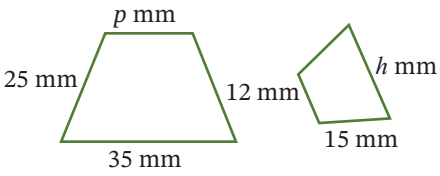
a



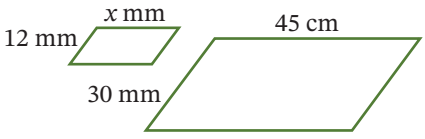
b



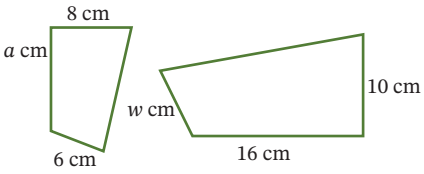
c



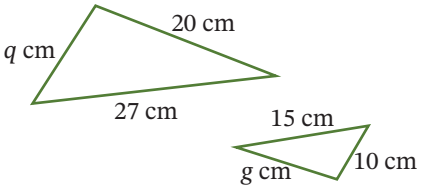
d



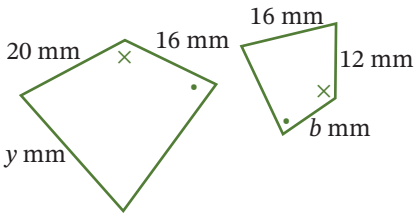
e



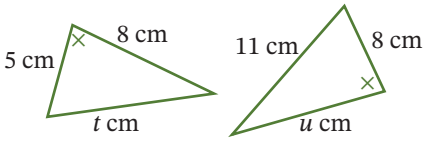
f



g

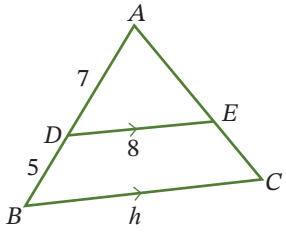


h

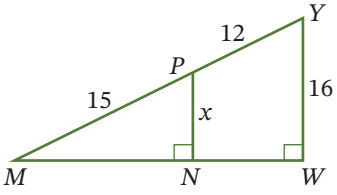


EXAMPLE 10

2 $\triangle ABC \sim \triangle ADE$. Find the value of h. R



3 $\triangle MNP \sim \triangle MWY$. Find the value of x. R



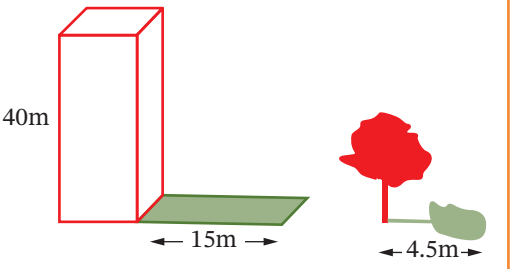
Foundation Standard Complex

4 This photograph of the Sydney Harbour Bridge has been enlarged so that its length is 24 cm. If the dimensions of the original photo were 15 cm × 10 cm, what is the width of the enlargement? R



Taki O/Shutterstock.com

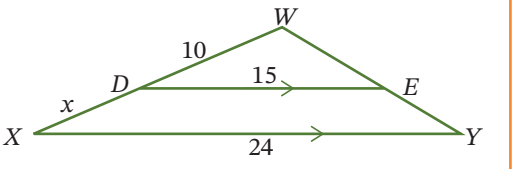
5 A building that is 40 m high casts a shadow 15 m long. At the same time, the shadow of a tree is 4.5 m long. What is the height of the tree? PS R



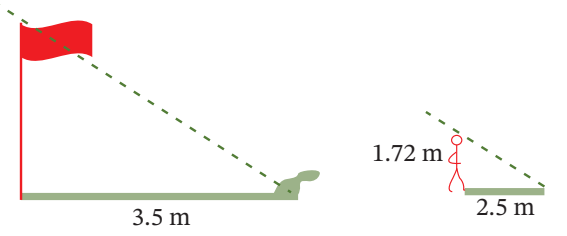
6 $\triangle WXY \sim \triangle WDE$. What is the value of x? Select the correct answer **A**, **B**, **C** or **D**. R

A 4 **B** 6

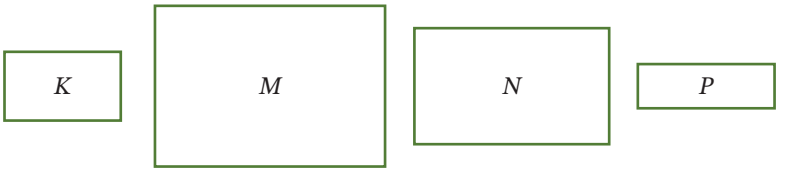
C 8 **D** 10



7 Katrina is 1.72 m tall and casts a shadow that is 2.5 m long. At the same time, a flagpole casts a shadow that is 3.5 m long. How long is the flagpole? PS R



8 Which 2 rectangles are similar? Select **A**, **B**, **C** or **D**.



A M and N **B** K and P **C** M and P **D** K and N

9 A 2 m high fence casts a shadow of 1.4 m. How long is the shadow cast by a pole that is 3.2 m high at the same time? PS R

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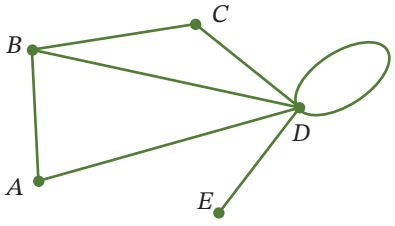
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Networks

10.05

A **network** is a collection of objects connected to each other in some way or system. Examples of networks include bus, rail and air routes, computer systems and family or friend relationships.

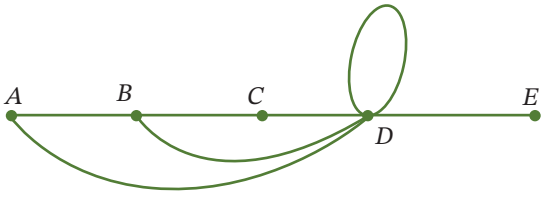
A **graph** is a diagram of a network where the objects are represented by points called **vertices** (or **nodes**), which are joined by lines called **edges**. The study of networks is called **graph theory**.



This graph or network diagram has 5 vertices, *A, B, C, D, E,* and 7 edges, *BC, BD, BA, AD, ED, CD, DD.*

A **loop** is an edge that connects a vertex to itself, such as *DD* in the diagram.

A network can be drawn in different ways as long as it shows the same connections. This diagram represents the same network as above.



PATH

Videos
Networks
The seven bridges of Konigsberg
Networks: Labyrinths and mazes

10.05

Degree of a vertex

The **degree** (or **order**) of a vertex is the number of edges connected to it, with loops counted as 2 edges. An **even vertex** has an even number of edges connected to it. An **odd vertex** has an odd number of edges.

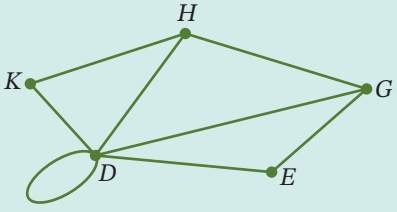
The **sum of the degrees** of all vertices in a network is equal to **twice the number of edges**, because each edge belongs to 2 vertices. This means that the sum of the degrees in a network is always even.

Example 11

a Find the degree of each vertex in this network.

b How many vertices are odd?

c Show that the sum of the degrees is equal to twice the number of edges.



Solution

a

Vertex	*D*	*E*	*G*	*H*	*K*
Degree	6	2	3	3	2

b There are 2 odd vertices.

c Number of edges = 8.

Sum of degrees of vertices = 6 + 2 + 3 + 3 + 2 = 16,

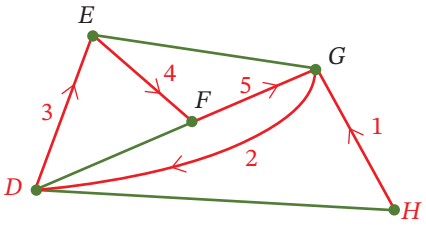
which is twice the number of edges (2 × 8).

PATH

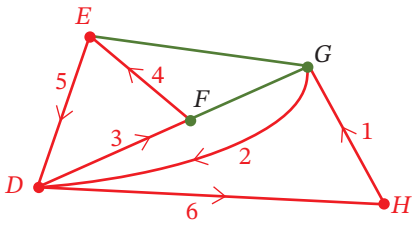
Walks, trails, paths and cycles

A **walk** is *any* route along the edges of a network, and may include repeated vertices or edges. Any walk that starts and ends at the same vertex is called a **closed walk**.

A **trail** is a walk with no repeated edges, such as shown in the network below. However, a vertex can be repeated, such as *G* in *HGDEFG*. A **circuit** is a **closed trail**, such as *HGDFEDH* shown below (starts and ends at *H*).



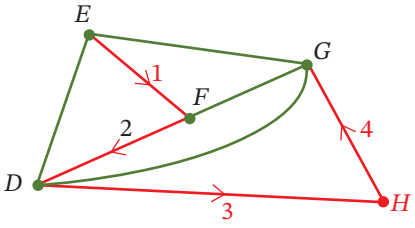
trail



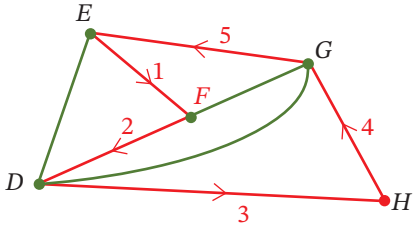
circuit

Think of a bushwalker walking on a trail or circuit and not repeating an edge.

A **path** is a walk with no repeated vertices (or edges), such as *EFDHG* shown in the network below. A **cycle** is a **closed path**, such as *EFDHGE* shown below (starts and ends at *E*, no repeated vertices).



path



cycle

Think of a cyclist riding along a bicycle path or cycle and not revisiting any points.

Example 12

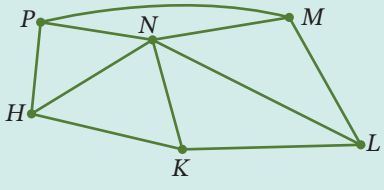
a State whether each walk is a trail, path or neither?

i *KLMPNHK* **ii** *PNMLNHK*

iii *HNMLNKHP* **iv** *LMNPHKNM*

b Write down a circuit that uses the edge *LN*.

c Write down a cycle the starts with *K* and uses the edge *HN*.



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Solution

a **i** *KLMPNHK* is a path since no vertices (and edges) are repeated.

ii *PNMLNHK* is a trail as no edges are repeated but vertex *N* is repeated.

iii *HNMLNKHP* is a trail as no edges are repeated but the vertices *N* and *H* are repeated.

iv *LMNPHKNM* is neither as the edge *MN* is repeated.

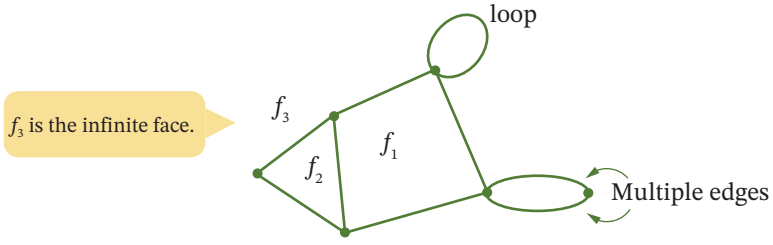
b *KLNMPHK* is a circuit because it starts and ends at *K* with no repeated edges.

c *KHNPMLK* is a cycle because it has no repeated vertices.

Faces of a network

A **face** of a network is a region bounded by edges. Regions formed by loops or multiple edges between the same vertices are *not* faces, but the external (outside) region of a network is a face. The external face is called the **infinite face**.

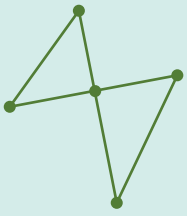
This network has 3 faces, f_1, f_2 and f_3.



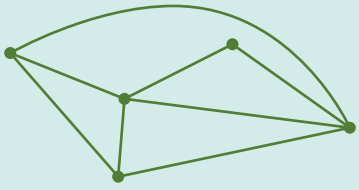
Example 13

Count the number of faces in each network.

a

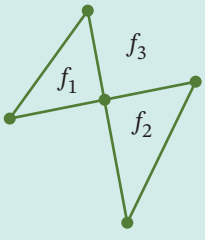


b



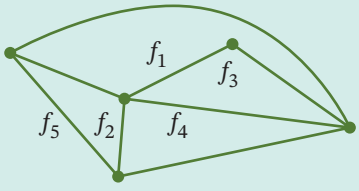
Solution

a



3 faces

b



5 faces

PATH

EXERCISE 10.05 ANSWERS ON P. 626

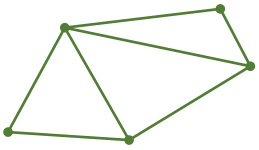
Networks U F R C

EXAMPLE 11

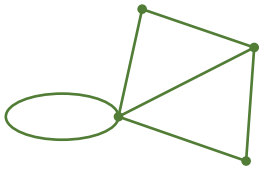
1 For each network, count the number of:

i vertices **ii** edges.

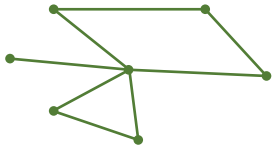
a



b



c

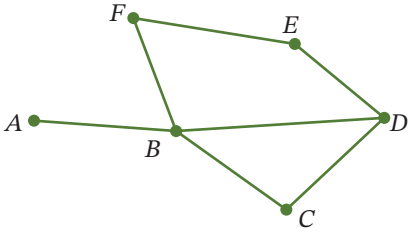


2 For each network: R C

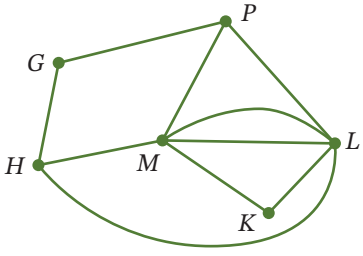
i find the degree of each vertex

ii show that the sum of the degrees is equal to twice the number of edges.

a

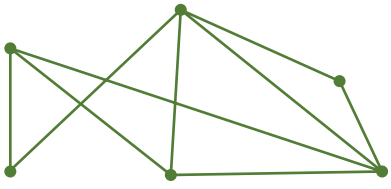


b

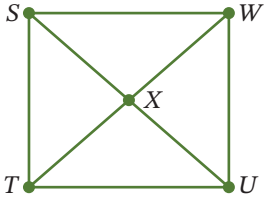


3 What is the sum of the degrees of all vertices in this graph? Select the correct answer **A**, **B**, **C** or **D**.

A 9 **B** 14 **C** 18 **D** 22

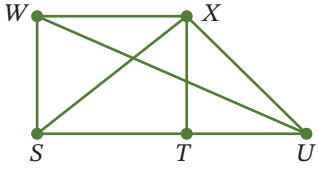


4 Which of these graphs shows the same information as the given network diagram?



Select **A**, **B**, **C** or **D**. R C

A



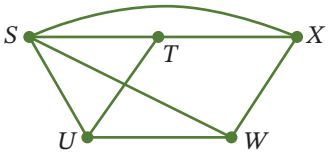
B



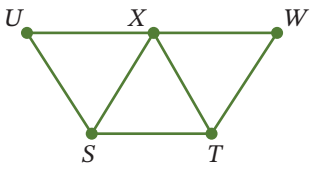
Foundation Standard Complex

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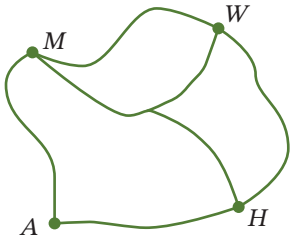
C



D

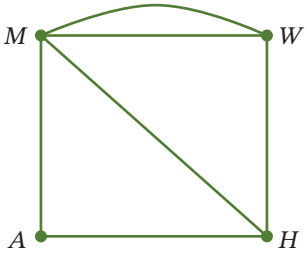


5 Which network diagram represents the map showing the roads that connect 4 towns?

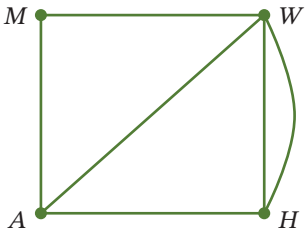


Select **A**, **B**, **C** or **D**. R C

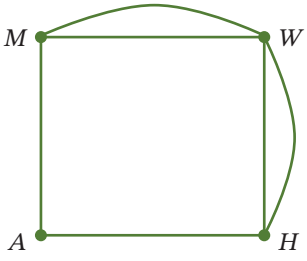
A



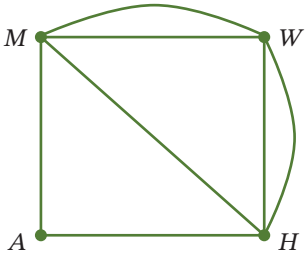
B



C



D



6 Anniek, Belinda, Emily, Mehek and Rima work at the same *J-Mart* store. R C

- Anniek is friends with Belinda, Emily and Rima
- Belinda is friends with Anniek and Emily
- Mehek is friends with Emily and Rima
- Rima is friends with Anniek, Mehek and Emily
- Emily is friends with Anniek, Belinda, Mehek and Rima

a Draw a graph to represent this network of friends.

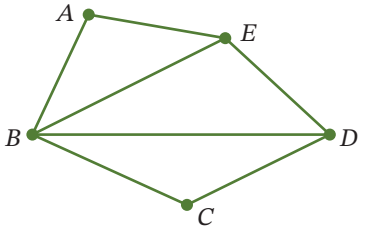
b Who is not a friend with Anniek?

Foundation Standard Complex

PATH

EXAMPLE 12

7 **a** State whether each walk is a trail or path. R C



i *AEBDCB* **ii** *EBDCBAE* **iii** *EDCBAE*

iv *BDCBAED* **v** *DBEABCD* **vi** *DCBEA*

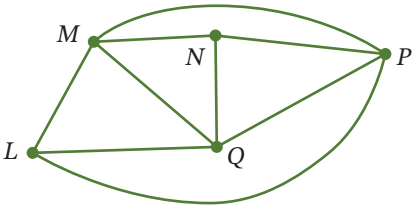
b Which of the walks in part **a** is:

i a circuit (a closed trail)? **ii** a cycle (closed path)?

c Write down a circuit that contain the edges *BC* and *DE*.

d Write down a cycle that contains the edge *CD*.

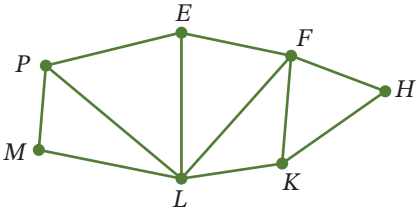
8 Which of the following is a circuit?



Select **A**, **B**, **C** or **D**. R C

A *MPQNMLPM* **B** *PMNPQLP* **C** *QNPMLPN* **D** *LQPLMQP*

9 **a** Which of the following walks is a cycle?



Select **A**, **B**, **C** or **D**. R C

A *LKHFLEPM* **B** *FHEKHFEM* **C** *MLKHFEKM* **D** *EFHKLPE*

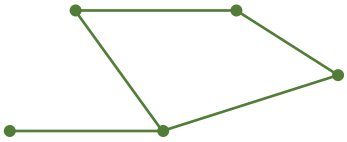
b Write down a cycle that begins at *K* and includes edge *LP*.

Foundation Standard Complex

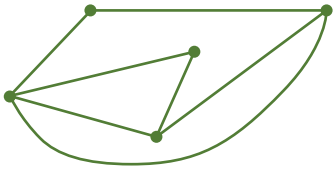
9780170479622

10 Count the number of faces in each network.

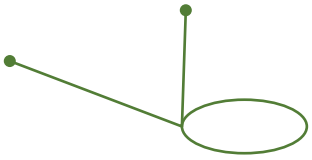
a



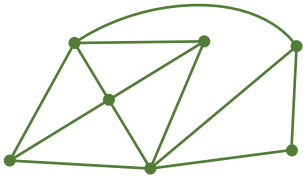
b



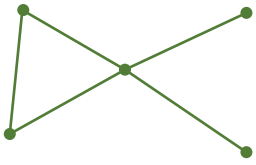
c



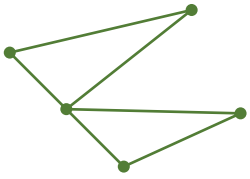
d



e



f



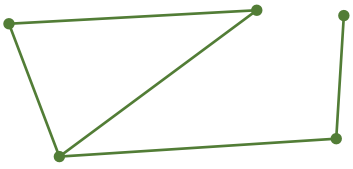
PATH

EXAMPLE 13

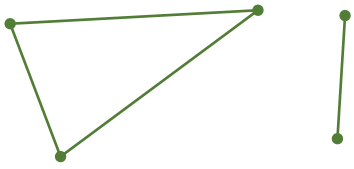
10.06

Planar graphs and Euler's formula 10.06

A network or graph is **connected** if there is a path between any 2 vertices on the network.

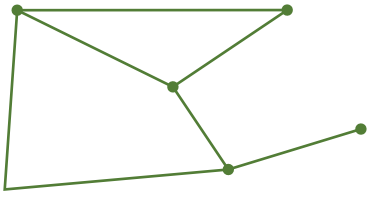


Connected

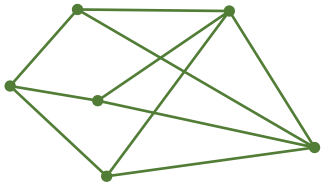


Not connected

A **planar graph** is a connected graph that can be drawn so that no edges cross over. 'Planar' means 2D, on a flat plane, and a planar graph can be drawn on a flat plane so that its edges don't overlap.



Planar graph



A non-planar graph

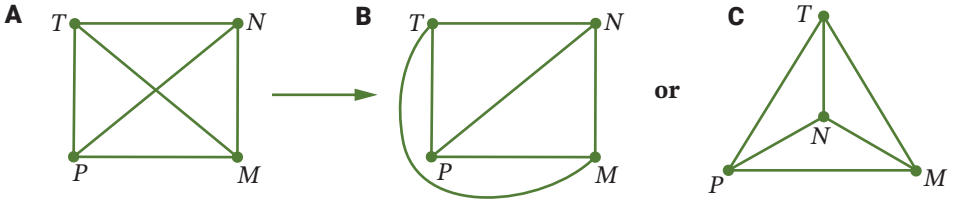
PATH

Video Euler's formula for planar graphs

Foundation | Standard | Complex

PATH

Some graphs may have edges crossing but they are still planar graphs because they can be *redrawn* without any of the edges crossing. For example, Graph **A** below is planar because it can be redrawn as **B** or **C**.

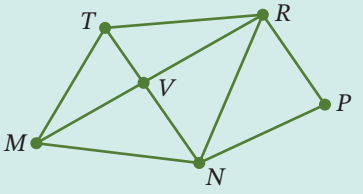


A **non-planar graph** has edges crossing and cannot be drawn so that the edges do not cross.

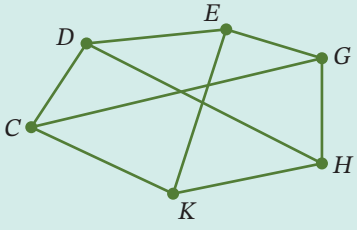
Example 14

State whether each network is a planar or non-planar graph.

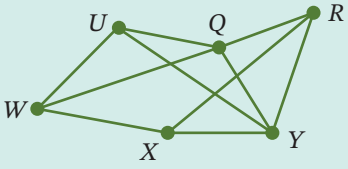
a



b



c

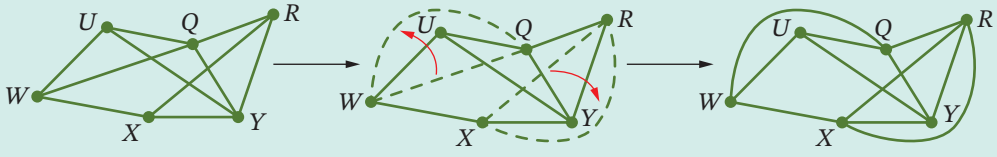


Solution

a A planar graph because no edges cross.

b It is impossible to move the edges so that no edges crossover so this is a non-planar graph.

c This graph can be redrawn by moving XR and WQ as shown.

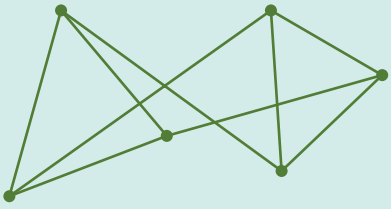


There are now no edges crossing so it's a planar graph.

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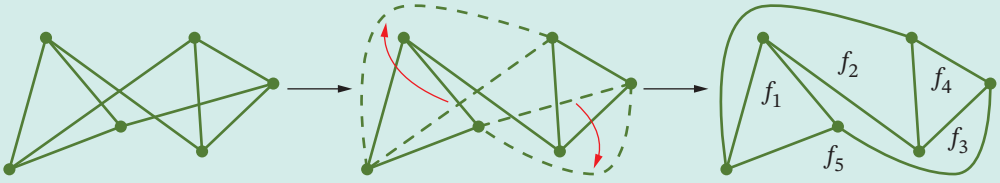
Example 15

Count the number of faces in this planar graph.



Solution

Redraw the graph so that no edges cross.



There are 5 faces.

Euler's formula

The Swiss mathematician Leonhard Euler (pronounced 'oiler') discovered a relationship between the vertices, edges and faces of a planar graph.

Euler's formula for planar graphs

For a graph to be planar,

$$v - e + f = 2$$

where

v = number of vertices

e = number of edges

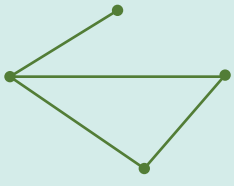
f = number of faces

Example 16

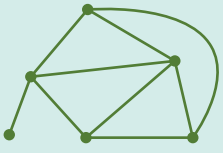
'Verify' means to show that something is true or correct.

Verify Euler's formula for each planar graph.

a



b



PATH

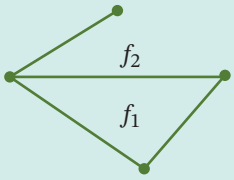
Solution

a $v = 4, e = 4$ and $f = 2$

$\therefore v - e + f = 4 - 4 + 2$

$= 2$

$\therefore$ Euler's formula is true.

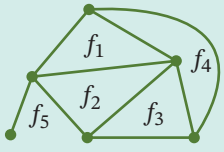


b $v = 6, e = 9$ and $f = 5$

$\therefore v - e + f = 6 - 9 + 5$

$= 2$

$\therefore$ Euler's formula is true.

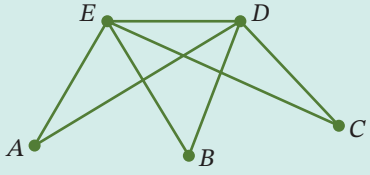


Example 17

Show that the graph is planar by:

a redrawing the graph so no edges cross

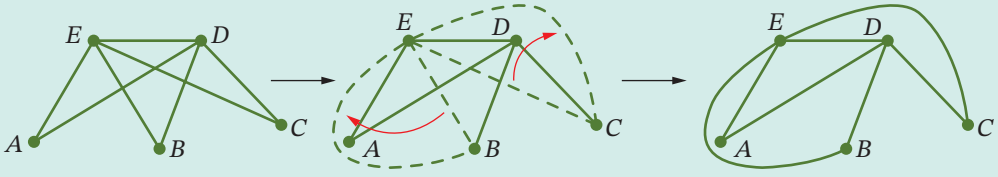
b using Euler's formula.



Solution

a Edge EC crosses edges AD and BD, so move edge EC to the outside of vertex D.
Also, edge EB crosses edge AD so move edge EB to the outside of vertex A.

Since the graph can be redrawn without any edges crossing, it is a planar graph

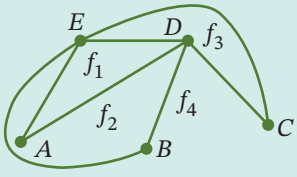


b $v = 5, e = 7$ and $f = 4$

$\therefore v - e + f = 5 - 7 + 4$

$= 2$

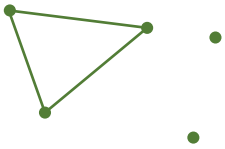
$\therefore$ the graph is planar

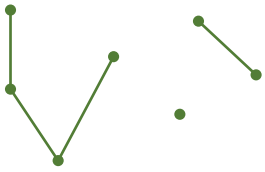


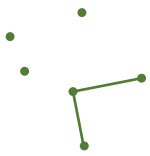
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Planar graphs and Euler's formula U F R C

1 What is the smallest number of edges required to make each graph a connected network. R C

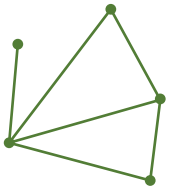
a 

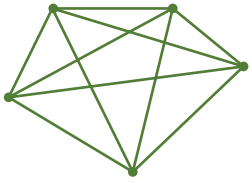
b 

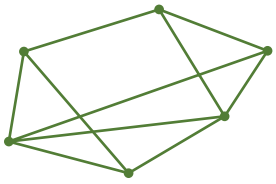
c 

EXAMPLE 14

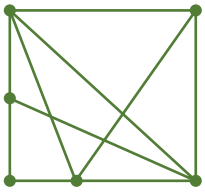
2 State whether each network is a planar or not planar graph. Give a reason for your answer. C

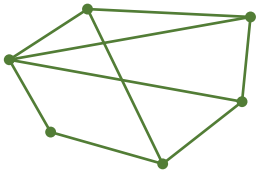
a 

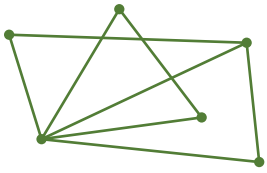
b 

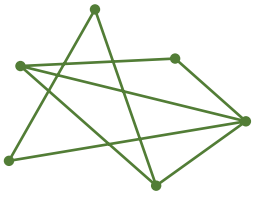
c 

3 Redraw each graph to show that it is planar. C

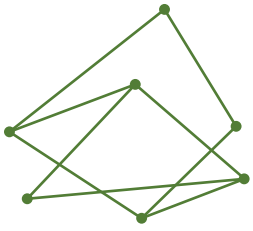
a 

b 

c 

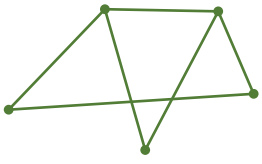
d 

e 

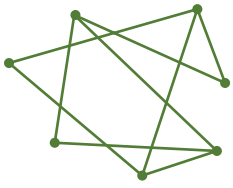
f 

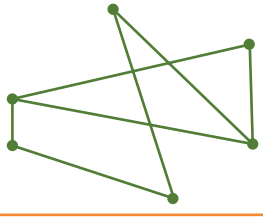
EXAMPLE 15

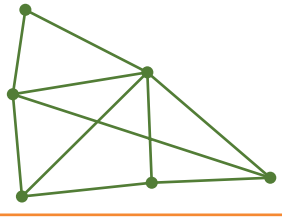
4 Redraw each planar graph to count its number of faces. R C

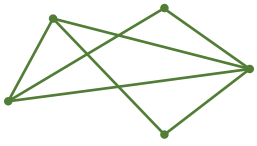
a 

b 

c 

d 

e 

f 

Foundation | Standard | Complex

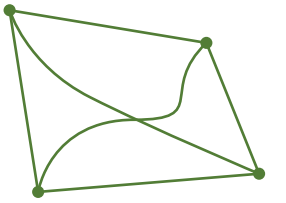
PATH

5 How many faces does this graph have?

Select the correct answer **A**, **B**, **C** or **D**.

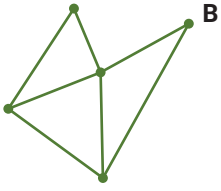
A 2 **B** 3

C 4 **D** 5

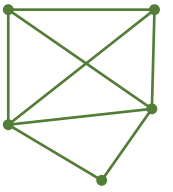


6 Which planar graph has 5 vertices and 5 faces? Select **A**, **B**, **C** or **D**.

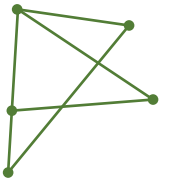
A



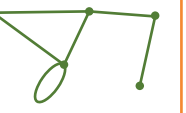
B



C



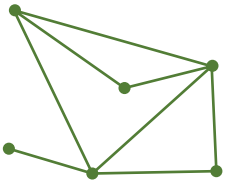
D



EXAMPLE 16

7 Verify Euler's formula for each planar graph. R C

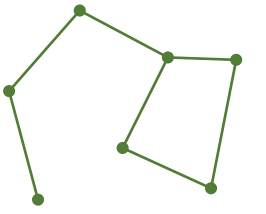
a



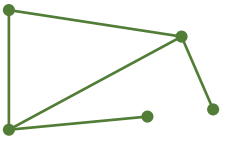
b



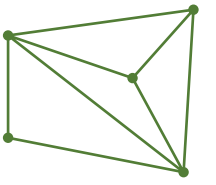
c



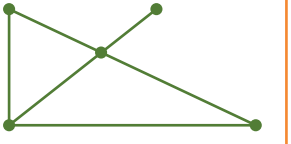
d



e



f

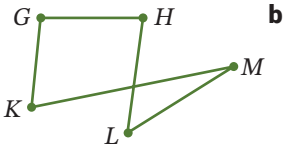


EXAMPLE 17

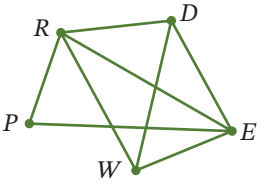
8 Show that each graph is planar by: R C

i redrawing the graph so no edges cross **ii** using Euler's formula.

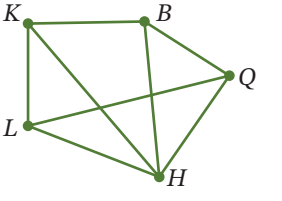
a



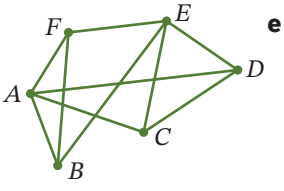
b



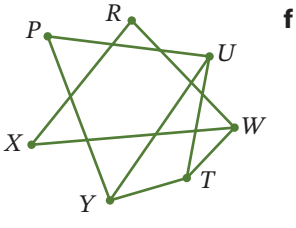
c



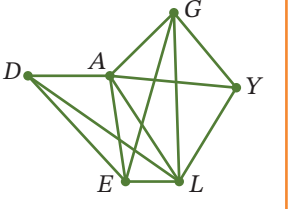
d



e



f



Foundation Standard Complex

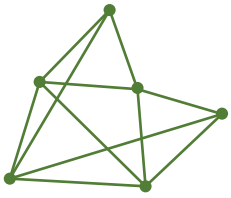
9780170479622

9 For which graph does Euler's formula NOT apply? Select **A**, **B**, **C** or **D**.

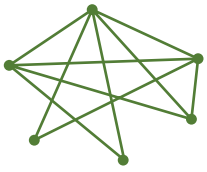
A



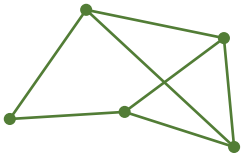
B



C



D



10 A planar graph has 7 vertices and 7 faces. What is its number of edges? Select **A**, **B**, **C** or **D**.

A 9 **B** 12 **C** 14 **D** 15

11 A planar graph has 10 faces and 13 edges. How many vertices has it? Select **A**, **B**, **C** or **D**.

A 3 **B** 5 **C** 8 **D** 11

12 Draw a planar network with 5 vertices and 7 faces.

13 A planar graph has 10 edges. Which of the following could be the number of vertices and faces? Select **A**, **B**, **C** or **D**.

A 9 vertices and 5 faces **B** 5 vertices and 5 faces

C 6 vertices and 7 faces **D** 8 vertices and 4 faces

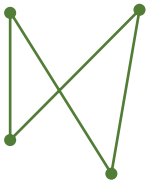
Investigation

Eulerian trails and circuits

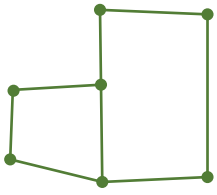
1 Copy each network, then trace over it with your pen or finger using each edge only once without lifting your pen or finger.

Check your work with that of other students.

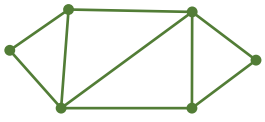
a



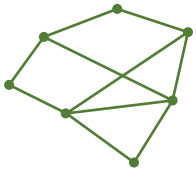
b



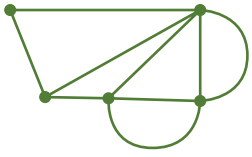
c



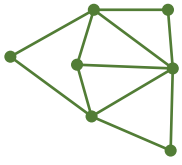
d



e



f



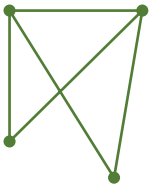
Foundation Standard Complex

9780170479622

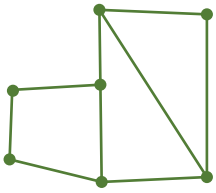
PATH

2 An extra edge has been added to each network in Question 1. Copy each network, then trace over it with your pen or finger using each edge only once. Which networks could you trace?

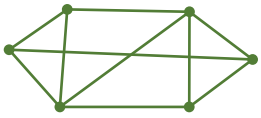
a



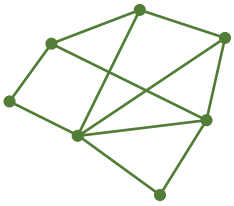
b



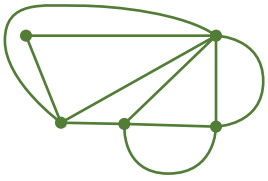
c



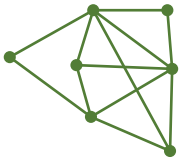
d



e



f



3 Copy and complete the table for each network in questions **1** and **2**.

Network	Number of odd vertices	Are all vertices even? (Y or N)	Is the network traceable? (Y or N)	Does trace begin and end at the same vertex? (Y or N)
1 a	0	Y	Y	Y
b				
c				
d				
e				
f				
2 a				
b				
c				
d				
e				
f				

4 Use the results in the table to copy and complete:

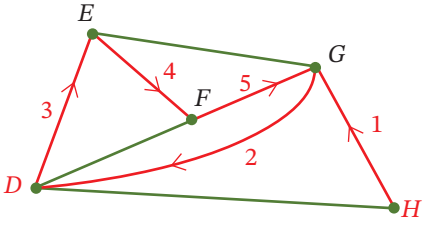
a A network can be traced if it is exactly _____ or _____ odd vertices.

b A network's trace starts and finishes at the same vertex if all vertices are _____.

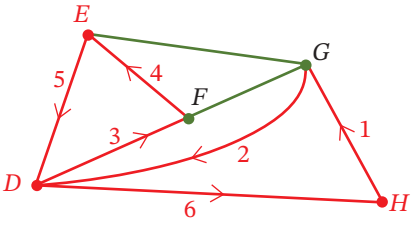
Eulerian trails and circuits

10.07

Remember that a **trail** is a walk in a network with no repeated edges, while a **circuit** is a **closed trail**, starting and ending at the same vertex.

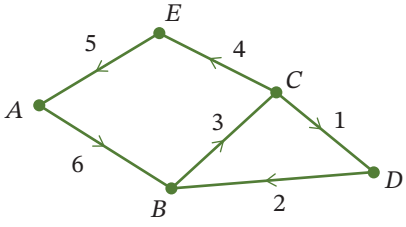


trail

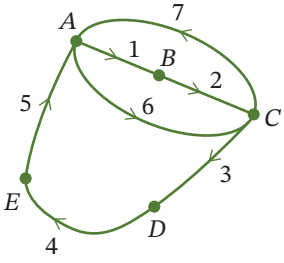


circuit

An **Eulerian trail** is a trail that uses *every* edge once, while an **Eulerian circuit** is a circuit that uses *every* edge once.



Eulerian trail



Eulerian circuit

Eulerian trails

An **Eulerian trail** uses every edge in the network only once.

For an Eulerian trail to exist, the network must have exactly 2 odd vertices.

Eulerian trails start at one odd vertex and finish at the other odd vertex.

PATH

Video Eulerian trails

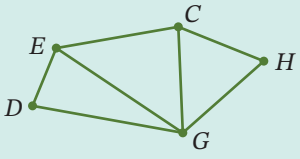
Worksheet Eulerian trails and circuits

PATH

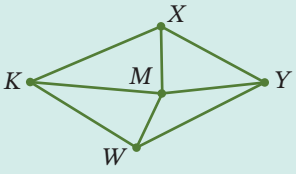
Example 18

Which network has an Eulerian trail?

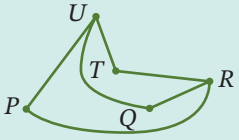
a



b



c



Solution

a This network has 2 odd vertices at C and E so it has an Eulerian trail, such as $EDGHCGEC$.

b This network has 4 odd vertices (degree 3) so it does not have an Eulerian trail.

c This network has 2 odd vertices at U and R (degree 3) so it has an Eulerian trail, such as $UPRQUTR$.

Eulerian circuits

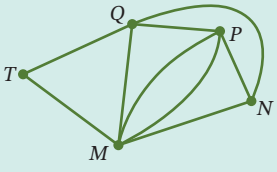
An **Eulerian circuit** uses every edge in the network only once and starts and ends at the same vertex.

For an Eulerian circuit to exist, all the vertices in the network must be even.

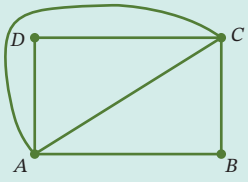
Example 19

Which network has an Eulerian circuit?

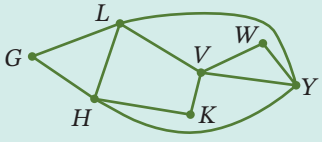
a



b



c



Solution

a Vertex M has degree 5 and vertex N has degree 3 so the network does not contain an Eulerian circuit.

b All vertices are even (B and D have degree 2, A and C have degree 4) so the network has an Eulerian circuit. Two possible Eulerian circuits are $BCADCAB$ and $ABCACDA$.

c All vertices are even (vertices G, K and W have degree 2, vertices H, L, V and Y have degree 4) so the network has an Eulerian circuit.

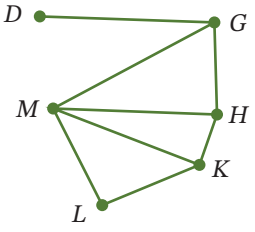
Eulerian trails and circuits U F R C

EXAMPLE 18

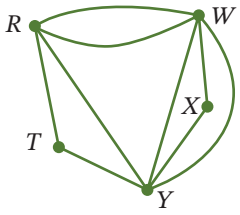
10.07

1 Which network has an Eulerian trail? Give reasons.

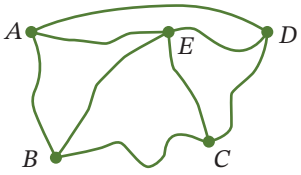
a



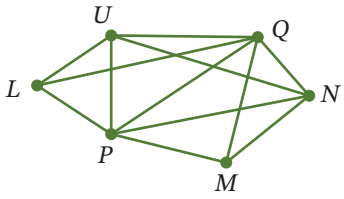
b



c



d



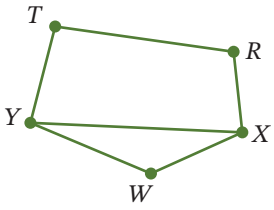
2 For each network: R C

i find the degree of each vertex

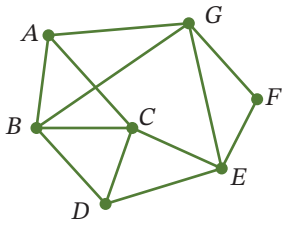
ii explain why the graph has an Eulerian trail

iii copy the graph and show a Eulerian trail on it.

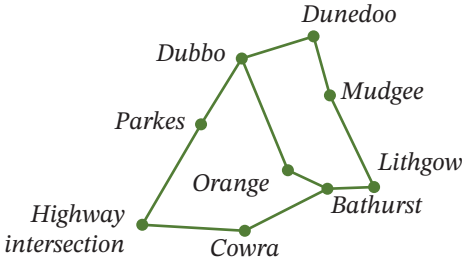
a



b



3 This graph shows the major roads linking cities in Central West NSW.



a Does this road network contain an Eulerian trail? Give reasons.

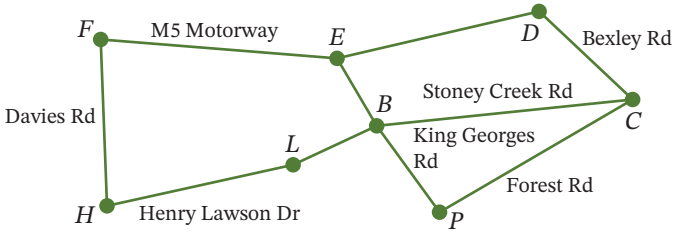
b Where will the trail start and finish?

c What is a possible Eulerian trail?

Foundation Standard Complex

PATH

4 This graph shows a network of major roads in the St George area, south of Sydney. R C

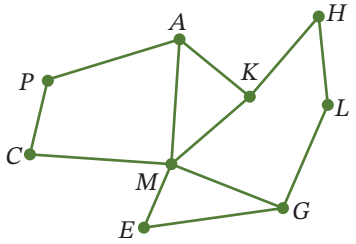


a Explain why the graph has an Eulerian trail.

b If the trail starts at E, where will it finish?

c Copy the graph and show the trail.

5 Which vertices can be joined by an edge so that the network contains an Eulerian trail?



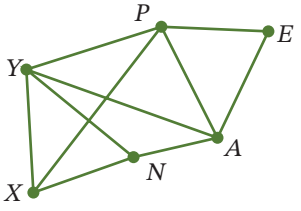
Select the correct answer **A**, **B**, **C** or **D**.

A G and K **B** M and P **C** G and H **D** K and L

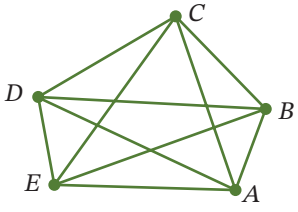
EXAMPLE 19

6 Test whether each network has an Eulerian circuit.

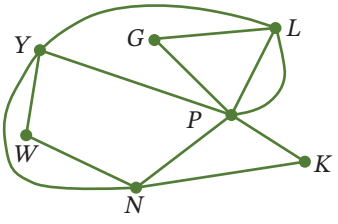
a



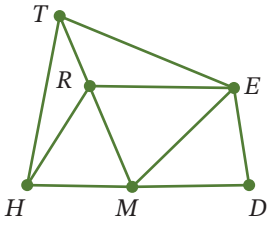
b



c



d



Foundation Standard Complex

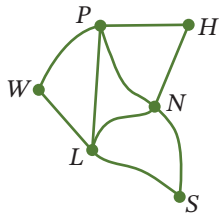
9780170479622

7 **a** Explain why this graph has an Eulerian circuit. R C

b Write down 2 Eulerian circuits.

c Copy the graph to draw the 2 circuits.

d What do you notice about your circuits?

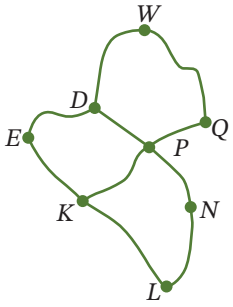


8 A road map for the delivery of milk to cafés and shops is shown by the network. The delivery starts at D and does not use the same road twice. R C

a Explain why the delivery cannot start and finish at D.

b A road is added to the network so that the delivery can start and finish at D. Which 2 vertices would the road join?

c Find 2 suitable routes for the milk delivery.



9 **a** This graph shows the roads connecting 4 towns.

Is it possible to start and finish at the same town after travelling along each road only once?

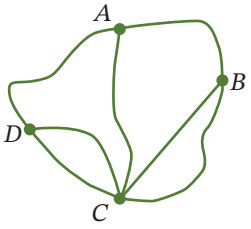
b A second road joining A to D is added.

Is it possible to travel along each road only once and

i start and finish at the same town?

ii start and finish at different towns?

Give reasons. R C



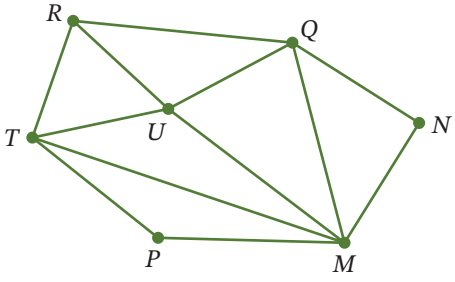
10 This tourist map shows 7 tourist destinations. Marinella and her family want to visit each destination, but use each road only once. R C

a From which destination should they start?

b At which destination will they finish their trip?

c How many destinations will Marinella and her family pass:

i only once? **ii** exactly twice?



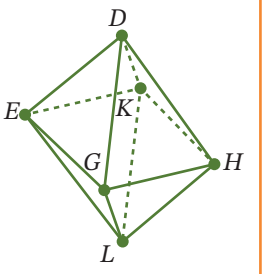
Foundation Standard Complex

PATH

11 An octahedron can be thought of as a 3D network.

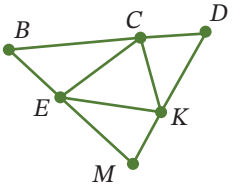
a Show that this network has an Eulerian circuit.

b Starting at D, describe a path that is an Eulerian circuit.

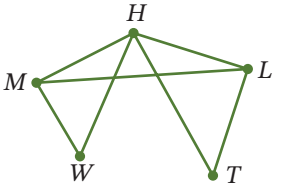


12 Determine which network is an Eulerian trail, an Eulerian circuit or neither. Give reasons for your answers. R C

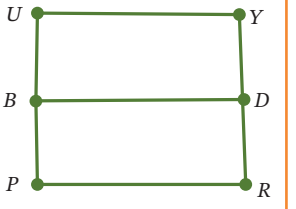
a



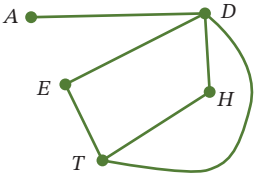
b



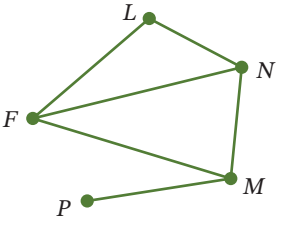
c



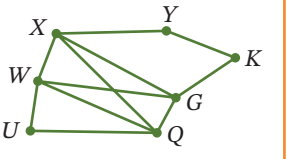
d



e

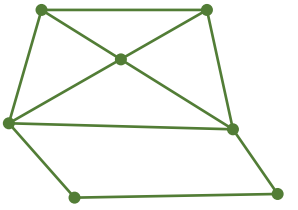


f

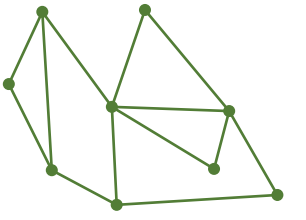


13 Which graph contains an Eulerian circuit? Select **A**, **B**, **C** or **D**.

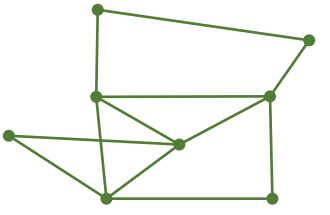
A



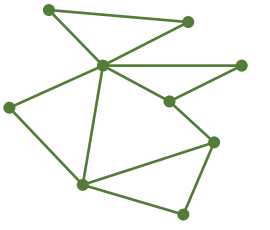
B



C



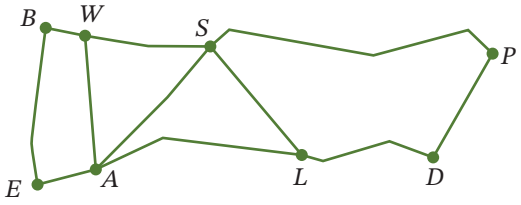
D



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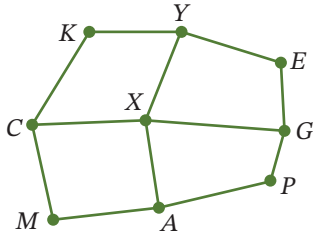
14 Mehek is competing in a 20 km road race and is given a map of the course as shown. To complete the race she must run along each road only once. R C



a At which point would the race start?

b Where would the race finish? Give reasons.

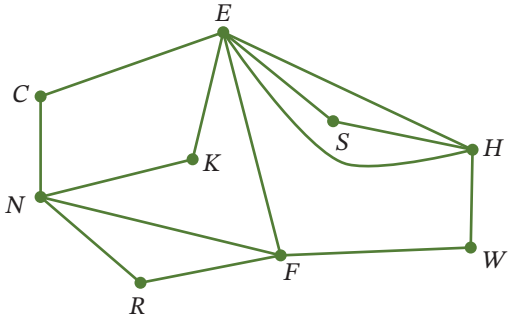
15 a Explain why this graph does not contain an Eulerian trail or circuit. R C



b How many vertices need to be joined for an Eulerian trail to exist in the graph?

c Which vertices should be joined for an Eulerian circuit to exist in the graph?

16 A marketing salesperson travels to stores in 8 different locations. In this network diagram, the vertices represent the locations and the edges represent the roads. R C



a Explain why the salesperson can travel from C, visit every location using each road only once and then return to location C.

b Describe the route the salesperson could take.

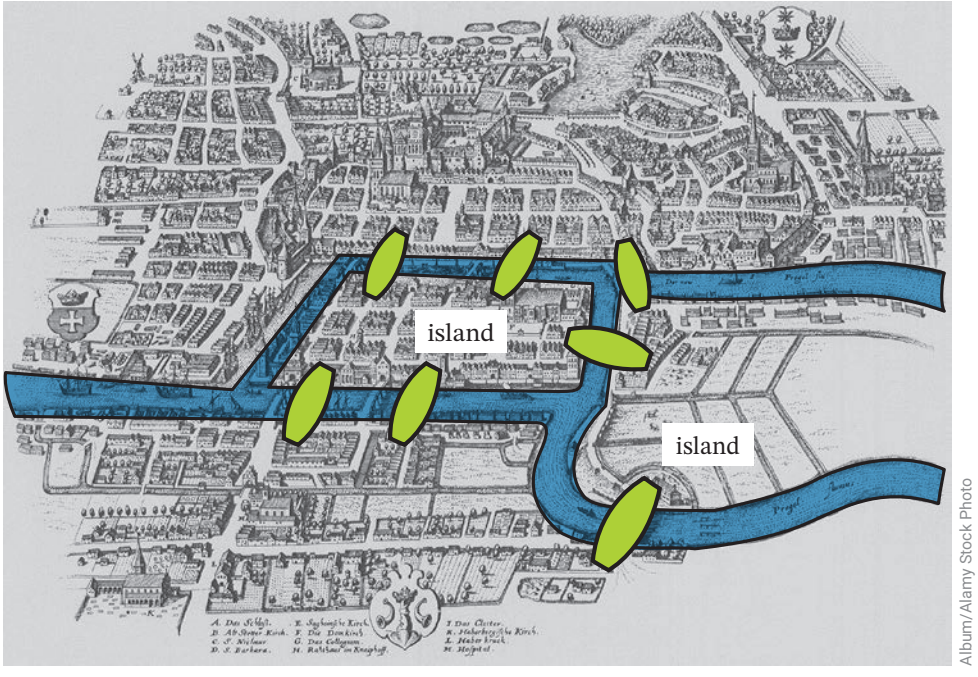
Foundation Standard Complex

Video
The seven bridges of Konigsberg

Did you know?

The 7 bridges of Königsberg

In the Prussian town of Königsberg (which is now Kaliningrad, Russia), there were 7 bridges connecting 2 islands and the banks of the Pregel River. There was a tradition where people would walk through the town on a Sunday and it became a challenge to complete a walk crossing each bridge only once. No person was able to do it but no one could prove that it was impossible.



Album/Alamy Stock Photo

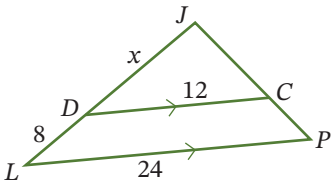
In 1736, the Swiss mathematician Leonard Euler solved the problem by reducing the map to a network. He found that it was impossible to do the walk. Eventually another bridge was built. That is why a walk that crosses every bridge exactly once is called an **Eulerian trail**. Euler developed a new branch of mathematics called **graph theory**, the geometry of networks.

1 **Draw a network to represent the map and explain why the original walk was impossible.**

2 **An eighth bridge was built. Where would the bridge have been placed so that the network could be traversed?**

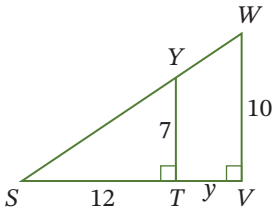
Power plus ANSWERS ON P. 628

1 $\triangle JDC \sim \triangle JLP$. Find the value of x.



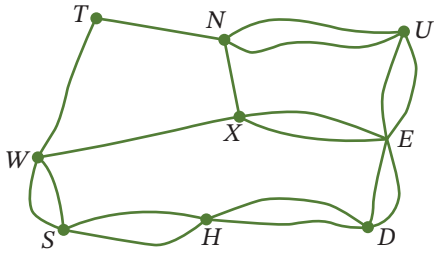
2 $\triangle STY \sim \triangle SVW$.

Find the value of y.



3 Leilani is surveying homes on family size.

Some streets have homes on both sides (shown by double edges on the network diagram) and other streets only have homes on one side only (shown by single edges).



a Explain why an Eulerian circuit can be found on this network.

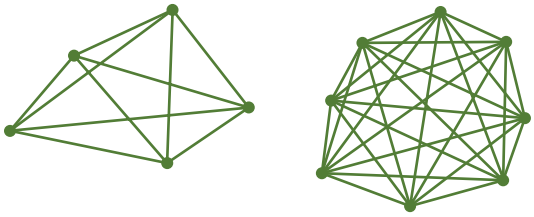
b Write down a circuit that begins at:

i N **ii** H **iii** X

c For each of your circuits from part **b**, write down which vertices occur:

i once in the circuit **ii** 3 times in the circuit.

4 In a polygon, each vertex is joined to all other vertices as shown (all diagonals are shown).



What type of polygon drawn this way will always have an Eulerian circuit?

CHAPTER 10 REVIEW

Quiz
Language of maths 10

Puzzle
Geometry and networks crossword

Language of maths

circuit	cycle	degree	edge
enlargement	Eulerian	face	image
line symmetry	matching	network	original
path	planar graph	reduction	reflection
rotation	rotational symmetry	scale factor	similar (~)
trail	transformation	translation	vertex (node)

1 What is **rotational symmetry**?

2 Describe what happens to a figure that is changed by a scale factor of $\frac{1}{2}$?

3 What is the symbol and meaning of 'is similar to'?

4 What type of graph can be drawn so that its edges don't cross?

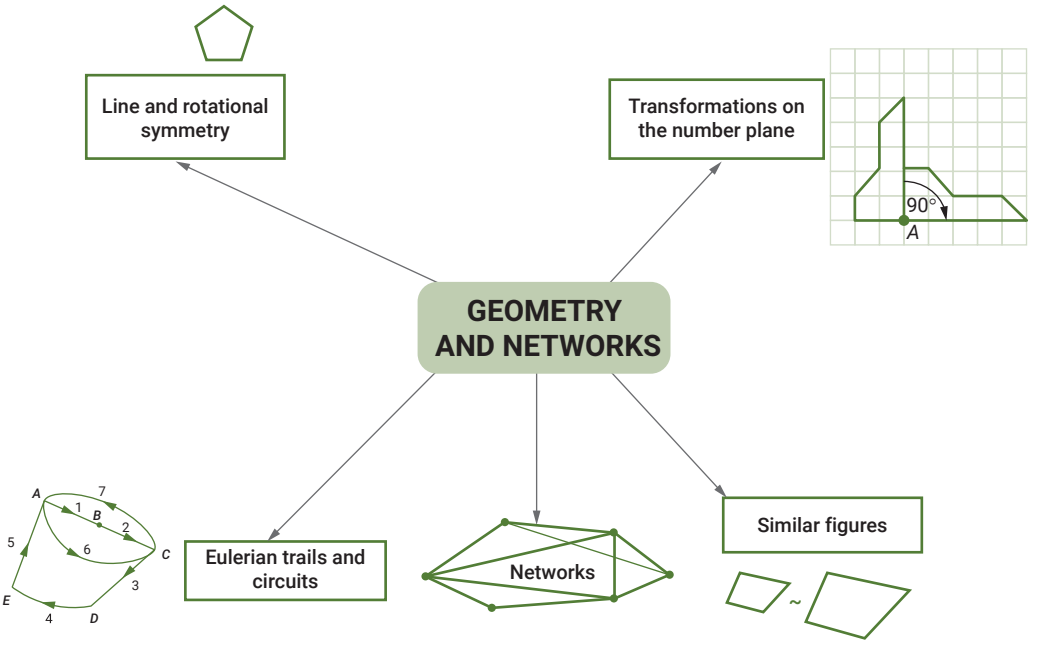
5 Explain what the formula $v - e + f = 2$ means.

6 Explain the difference between an **Eulerian trail** and an **Eulerian circuit**.

Worksheet
Mind map: Geometry and networks

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



9780170479622

TEST YOURSELF 10

ANSWERS ON P. 628

1 For each shape, find:

i the number of axes of symmetry

ii the order of rotational symmetry

a isosceles triangle **b** regular octagon **c** parallelogram.

2 **a** Copy the quadrilateral *PLMN* onto grid paper and then draw its image after:

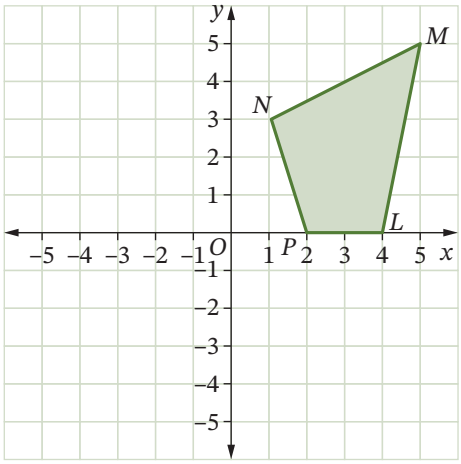
i reflecting it across the *y*-axis

ii reflecting it across the *x*-axis

iii rotating it anticlockwise about *O* through 90°

iv translating it 5 units to the left and 4 units down.

b For each of the above transformations, compare the coordinates of each vertex of the original to its matching vertex in its image.



3 Find the new position of the point $K(-5, 3)$ if it is:

a reflected in the *y*-axis

b reflected in the *x*-axis

c rotated clockwise about *O* through

i 90° **ii** 180° **iii** 270°

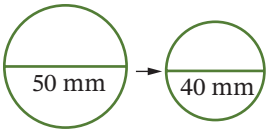
d translated 8 units to the right and 5 units down

e translated 3 units to the right, 2 units up and then rotated 180° about *O*.

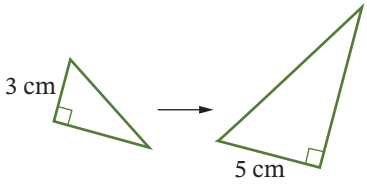
4 The point $W(-4, -2)$ is the image of a point after it has been reflected in the *x*-axis and then rotated clockwise about *O* through 90°. Find the original point.

5 Calculate the scale factor between each pair of similar figures.

a



b



PATH

Quiz
Test yourself 10

10.01

Worksheet
5 mm grid paper

10.02

10.02

10.02

10.03

TEST YOURSELF 10

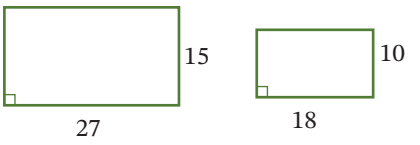
Foundation Standard Complex

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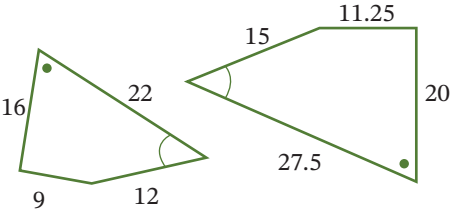
10.03

6 Test whether each pair of figures are similar.

a



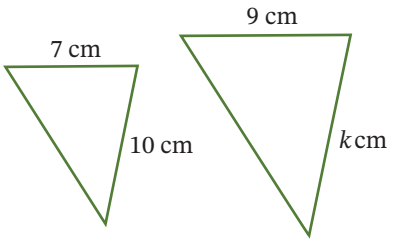
b



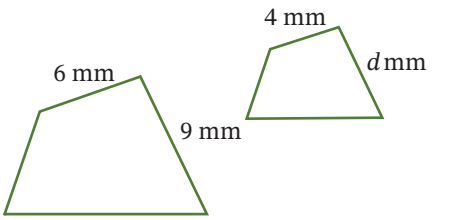
10.04

7 Find the value of the variable in each pair of similar figures.

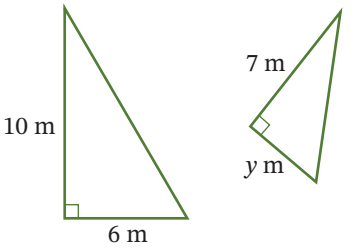
a



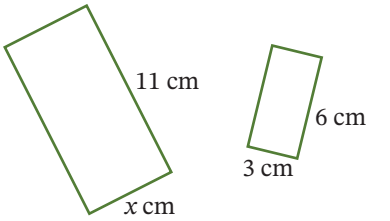
b



c

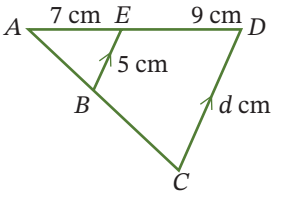


d



10.04

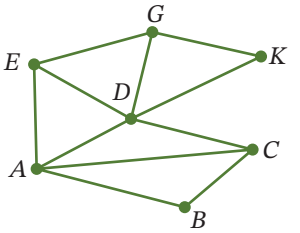
8 If $\triangle ABE \sim \triangle ACD$, find the value of d.



PATH

10.05

9



a State whether each walk is a trail or a path.

i *EABCDKG*

ii *CADKGEABC*

iii *EDKGDAB*

iv *DCBAEGKD*

b Which of the walks in part **a** is a **circuit**?

Foundation Standard Complex

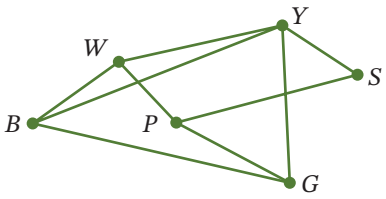
9780170479622

10 How many faces does this graph have?

Select **A**, **B**, **C** or **D**.

A 5 **B** 6

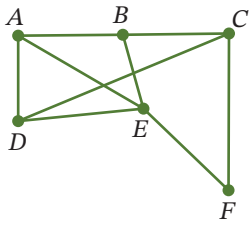
C 7 **D** 8



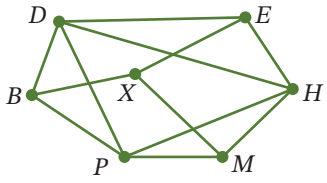
11 Show that the graph is planar by:

i redrawing it so that no edges cross **ii** using Euler's formula.

a



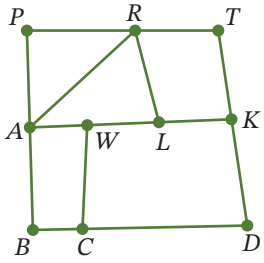
b



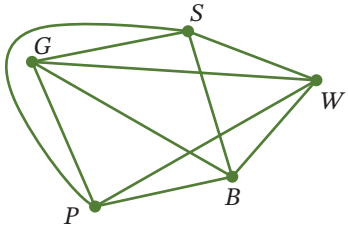
12 Determine whether each network has an Eulerian trail, an Eulerian circuit or neither.

Give a reason for your answer.

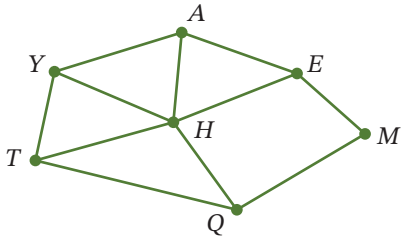
a



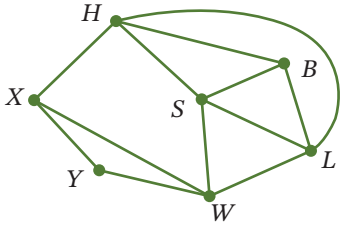
b



c



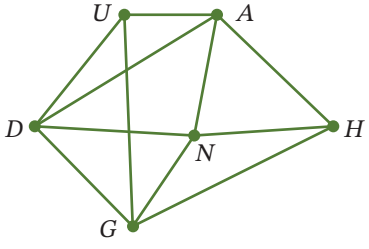
d



13 The graph shows how 6 towns are connected by a road network.

Is it possible to start at one town and go along each road once and finish at the same town?

Give a reason for your answer.



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